Swine Picornaviral Diseases

Chi-Fei Kao (高啓霏)

2026.10.01

Foot-and-mouth disease (FMD)

Introduction

- A **highly contagious**, **vesicular** disease of **cloven-hoofed** animals that can cause severe economic loss.
- FMD is on the World Organization for Animal Health **(OIE) list** of diseases.

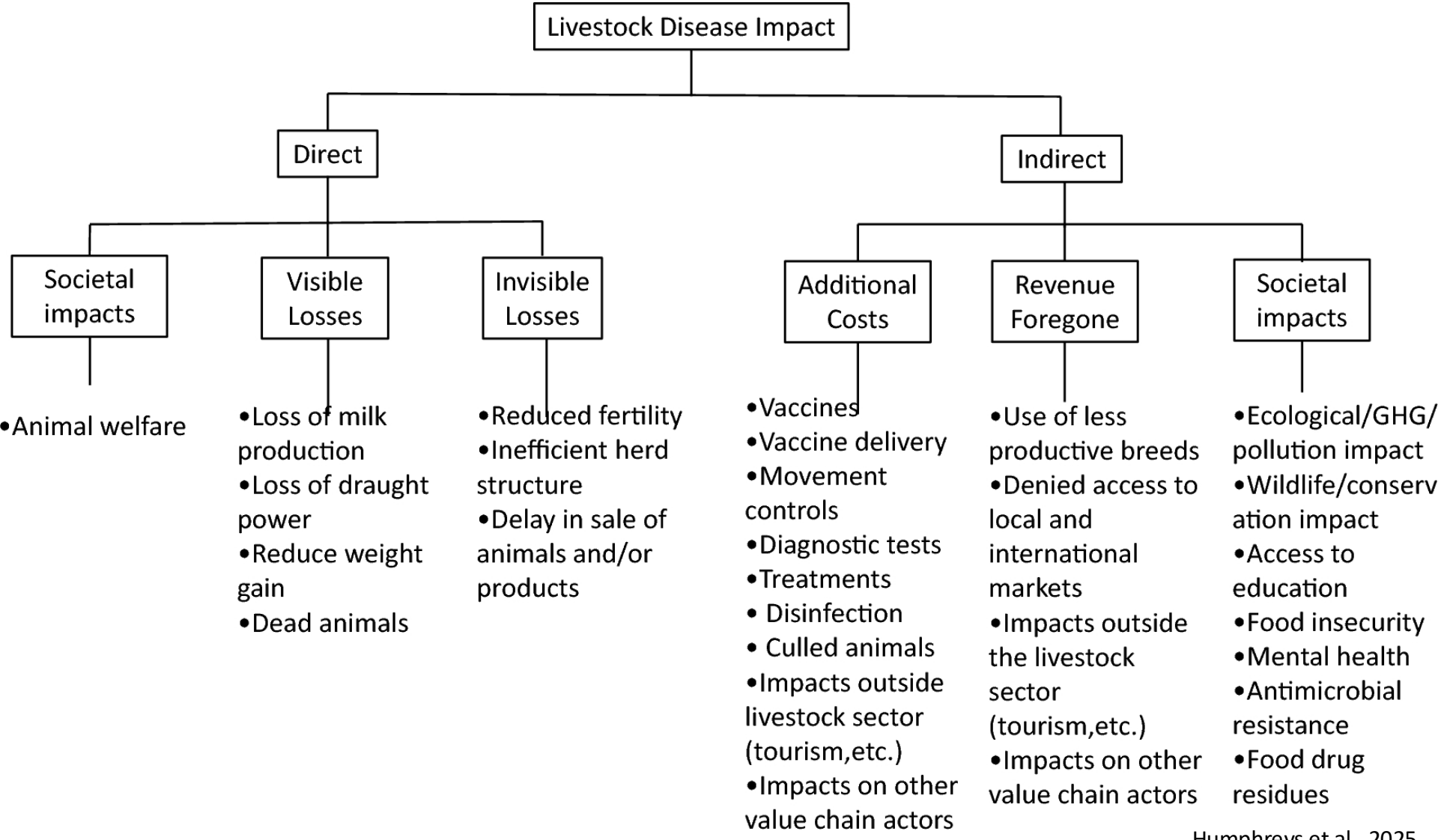

聯合晚報
UNITED EVENING NEWS
豬口蹄疫爆發 豬肉出口叫停
光復後，首度爆發！農委會宣布 9縣市20養豬場發現疫情 將緊急進口疫苗 連戰指示全力撲滅
疫情，嚴重；官員，凝重！
下月一日訪台
金瑞契亞洲行





https://epost.moa.gov.tw/theme_data.php?theme=epost&sub_theme=photo&id=910

Impact of FMD



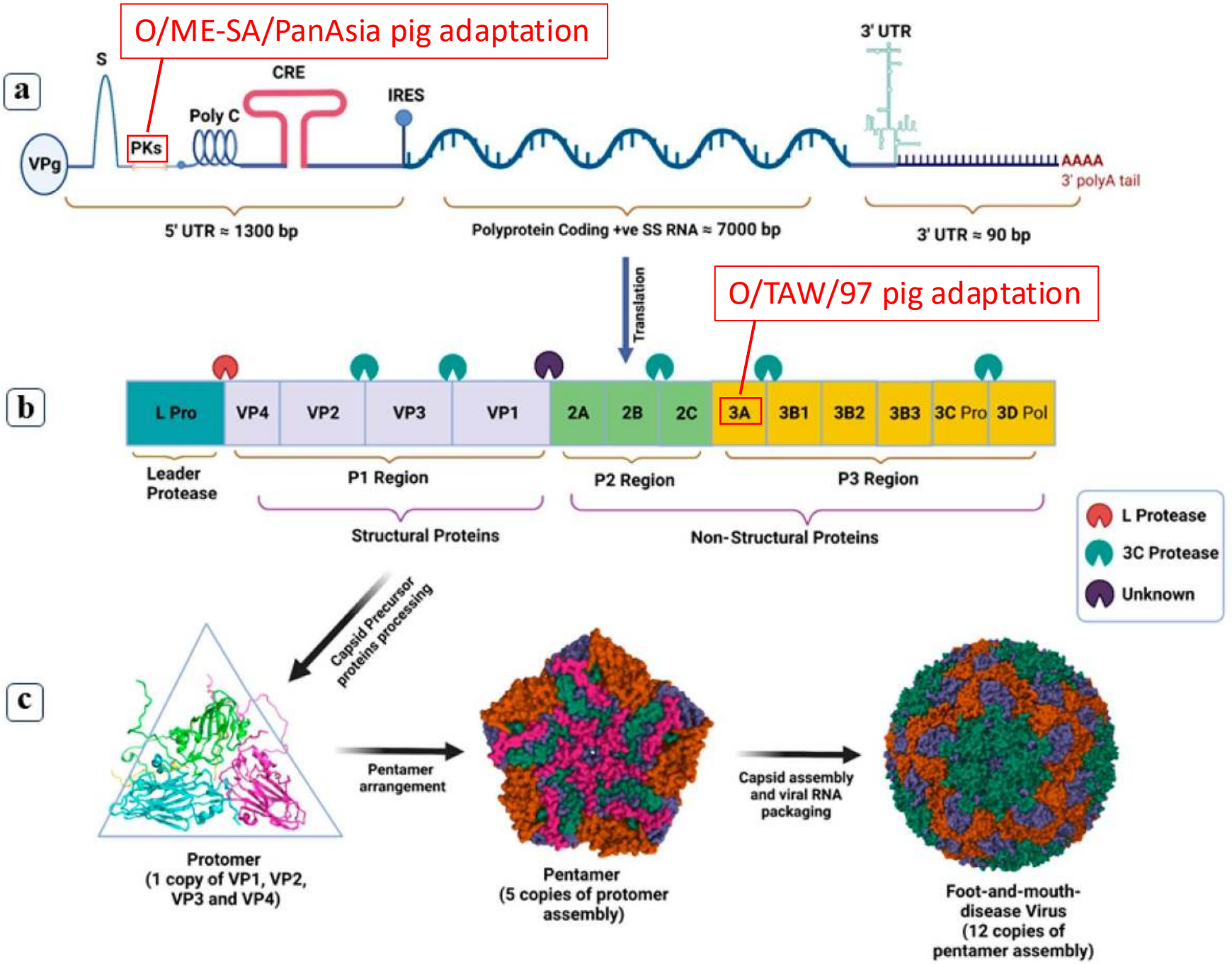
Humphreys et al., 2025

Etiology

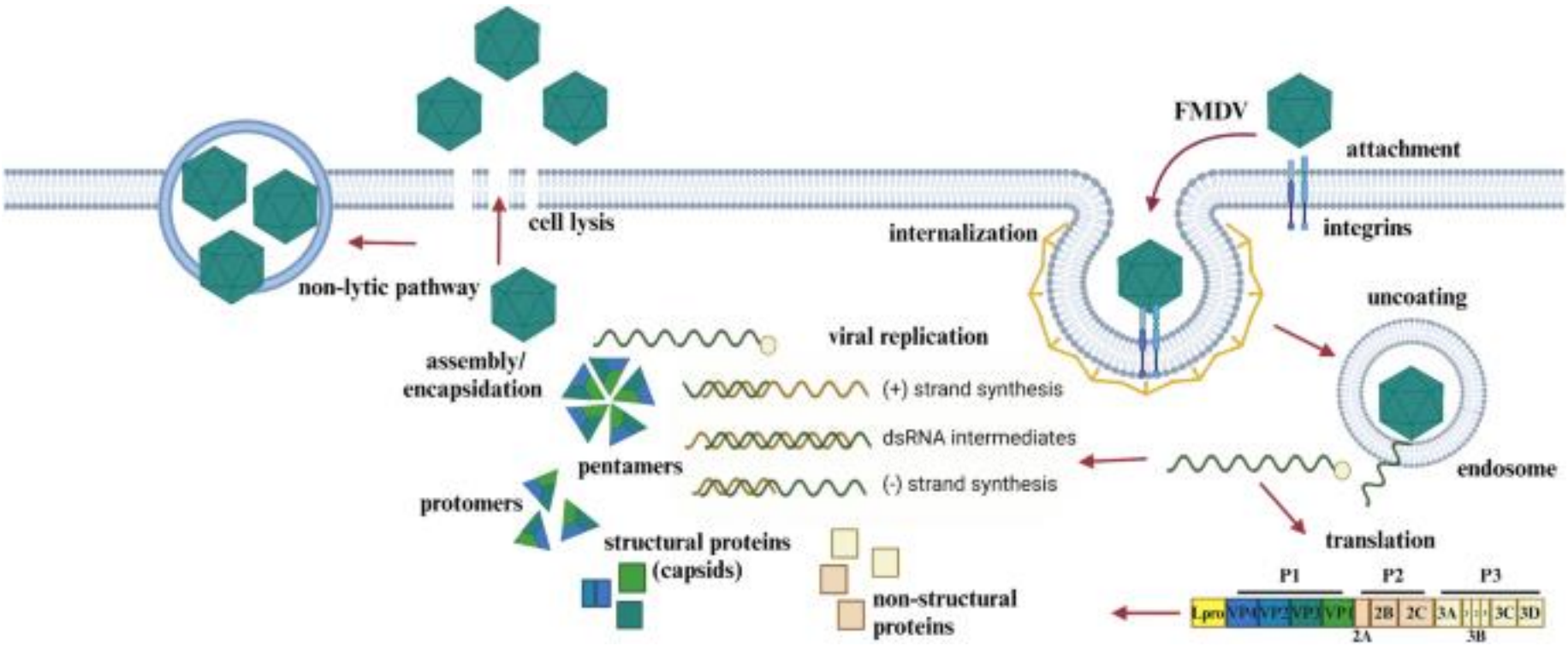
Foot-and-mouth disease virus (FMDV)

- Genus *Aphthovirus*, family of *Picornaviridae*
- **Non-enveloped** (VP1-4), icosahedral, 20-30 nm
- ss, positive-sense RNA genome, ~8.3 kb in length
- **Seven serotype**: A, O, C, SAT1, SAT2, SAT3, and Asia 1
- **Artiodactyla** (cloven-hoofed animals)
- Not zoonotic

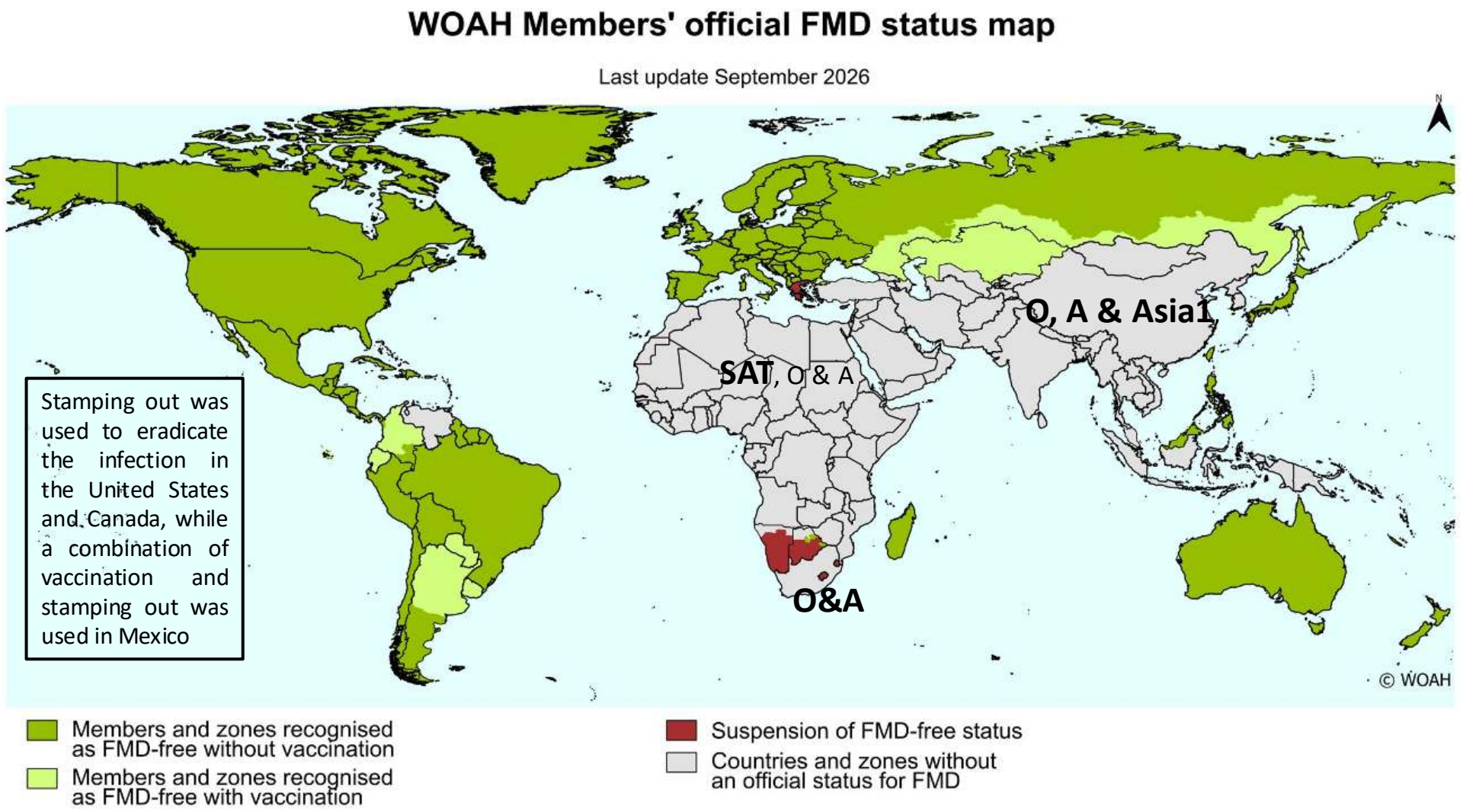
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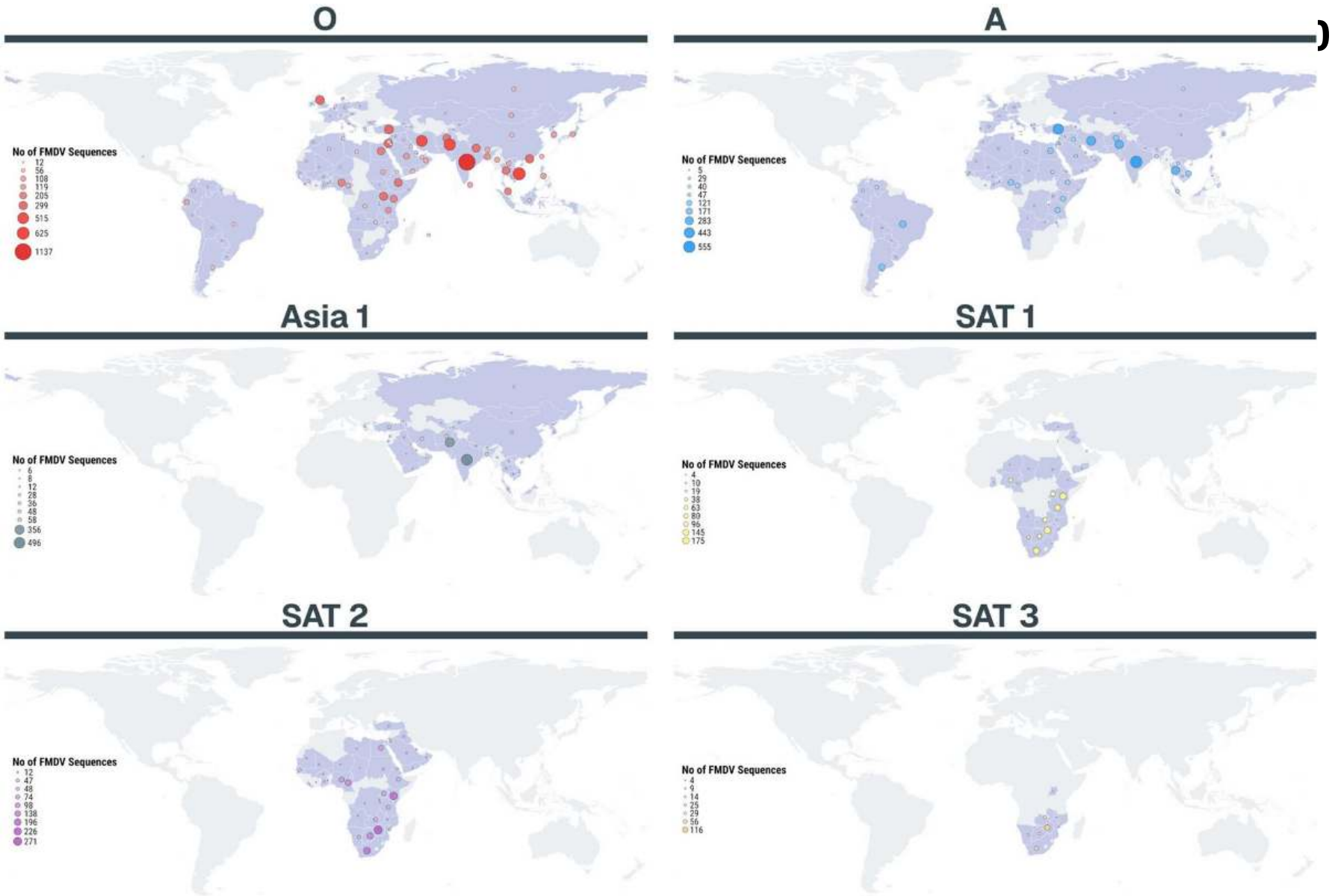


Mushtaq et al., 2024


FMDV
attachment
integrins
internalization
uncoating
endosome
translation
P1
P2
P3
Lpro
VP4
VP2
VP3
VP1
2A
2B
2C
3A
3B
3C
3D
viral replication
(+) strand synthesis
dsRNA intermediates
(-) strand synthesis
pentamers
protomers
assembly/
encapsidation
non-lytic pathway
cell lysis
structural proteins
(capsids)
non-structural
proteins

Epidemiology





Geographical distribution of FMDV sequence data that are available for the six circulating serotypes. Data shown represent sequences available at www.FMDbase.org. NB: data for serotype C are not shown since this serotype has **not been detected** anywhere globally since 2004.

Humphreys et al., 2025



我國口蹄疫疫情歷程

86年3月爆發口蹄疫疫情(O/Taiwan/97)(親豬型)

88年金門及臺灣發生口蹄疫案例(O/Taiwan/99株)(泛亞型)

92年5月獲認定為「施打疫苗之口蹄疫非疫國」

預計98年全面停止施打疫苗，惟同年2月起陸續發生7起口蹄疫案例，同年8月恢復施打

101年金門發生豬隻口蹄疫案例(O型SEA病毒株)(東南亞型)

102年5月臺中豬場最後1例口蹄疫0案例

104年金門牛發生2例A型口蹄疫(A型東南亞型)

106年5月OIE認定臺、澎、馬為施打疫苗口蹄疫非疫區

107年7月1日臺、澎、馬地區停止施打口蹄疫疫苗

https://www.biogas.com.tw/ReadFile/?p=Events&n=b3af3ea9-caa8-4681-a691-74523a6a8961.pdf

Transmission

- Susceptible animals

  -- **Of great significance**: **pigs**, **cattle**, **small ruminants** (goats and sheep) and water buffalo

  -- Some level of risk: elk, deer, llamas, alpacas, and Bactrian camels (雙峰駱駝), but not dromedary camels (單峰駱駝)

- Susceptible to experimental infection of FMDV

  -- Rat and hedgehog

  -- FMDV-induced myocarditis in young dogs that had scavenged FMDV-contaminated meat has been reported

Transmission

	Pig	Cattle	Sheep & goats
Role in epidemiology	**Amplifier**	**Indicator**	**Maintenance**
Minimal titer required to establish infection	10^3 $TCID_{50}$	**10 $TCID_{50}$**	15 $TCID_{50}$
Titer of aerosolized FMDV/animal/day	**10^6–$10^{8.6}$ $TCID_{50}$**	10^4–10^5 $TCID_{50}$	10^4–10^5 $TCID_{50}$
Duration of FMDV persistence	< 28 days	**Years**	months
Clinical symptoms	Fever, vesicles around the mouth and feet, lameness,	Fever, vesicles, drooling	**Subtle & various** ***Outbreak in the UK 2001**

The maximum reported duration of the carrier state in ruminants is species dependent: cattle, 3.5 years; sheep, 9 months; goats, 4 months; African buffalo, 5 years; and water buffalo, 2 months.

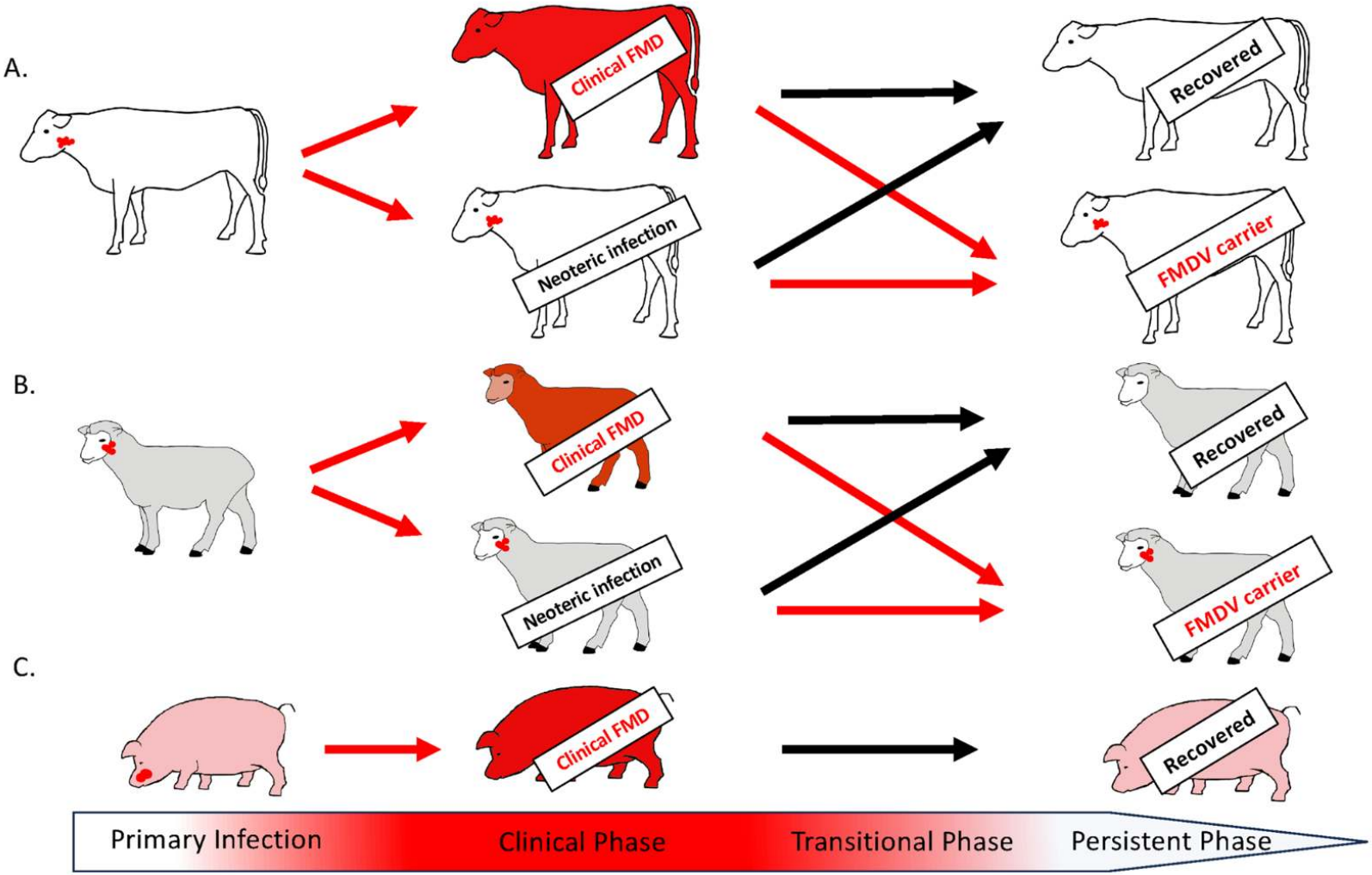


Figure 2 Schematic illustration of FMDV progression in different domestic livestock species. **A** In cattle, primary FMDV infection is localized to the nasopharynx (upper respiratory tract). Clinically susceptible cattle will progress through a phase of viremia and clinical disease, whereas animals protected by vaccination, previous exposure, or with breed-related resistance to clinical FMD will progress through a phase of neoteric infection during which infection is restricted to the nasopharynx, but virus shedding in oronasal secretions is common. Both clinically susceptible and protected cattle progress to either fully clear infection, or to maintain subclinical persistent FMDV infection of the upper respiratory tract through prolonged durations (the FMDV carrier state). **B** Sheep are infected via the respiratory tract (nasopharynx). The clinical severity of FMD varies greatly in sheep, and subclinical neoteric infections are common. Similar to cattle, sheep progress to either clear infection or become FMDV carriers. The site of virus persistence in sheep is localized to epithelial crypts of the oropharyngeal tonsils. **C** Pigs will primarily become infected via oral exposure to FMDV, with primary infection in the oropharynx, although the minimum infectious dose is higher than for cattle and sheep. Clinical signs of FMD in pigs are often dramatic, and infected animals shed very large quantities of virus. Pigs efficiently clear FMDV from all tissues following recovery from the clinical phase, and there is no FMDV carrier state in pigs.

Stenfeldt et al., 2025

Transmission

FMDV can be found in **all secretion and excretion** of acutely infected animals, including expired air, blisters...

- Direct contact (most common)
  - -- Nose-to-nose or skin contact **pigs**
  - -- Ingestion of contaminated/uncooked food (swill feeding)
  - -- FMDV-containing aerosol (less than 20 km) **Pig to cattle**
- Indirect
  - -- FMDV-contaminated personnel, vehicles, products... **pigs**

Spreading:

- People
- Equipment
- Vehicle
- Fomite

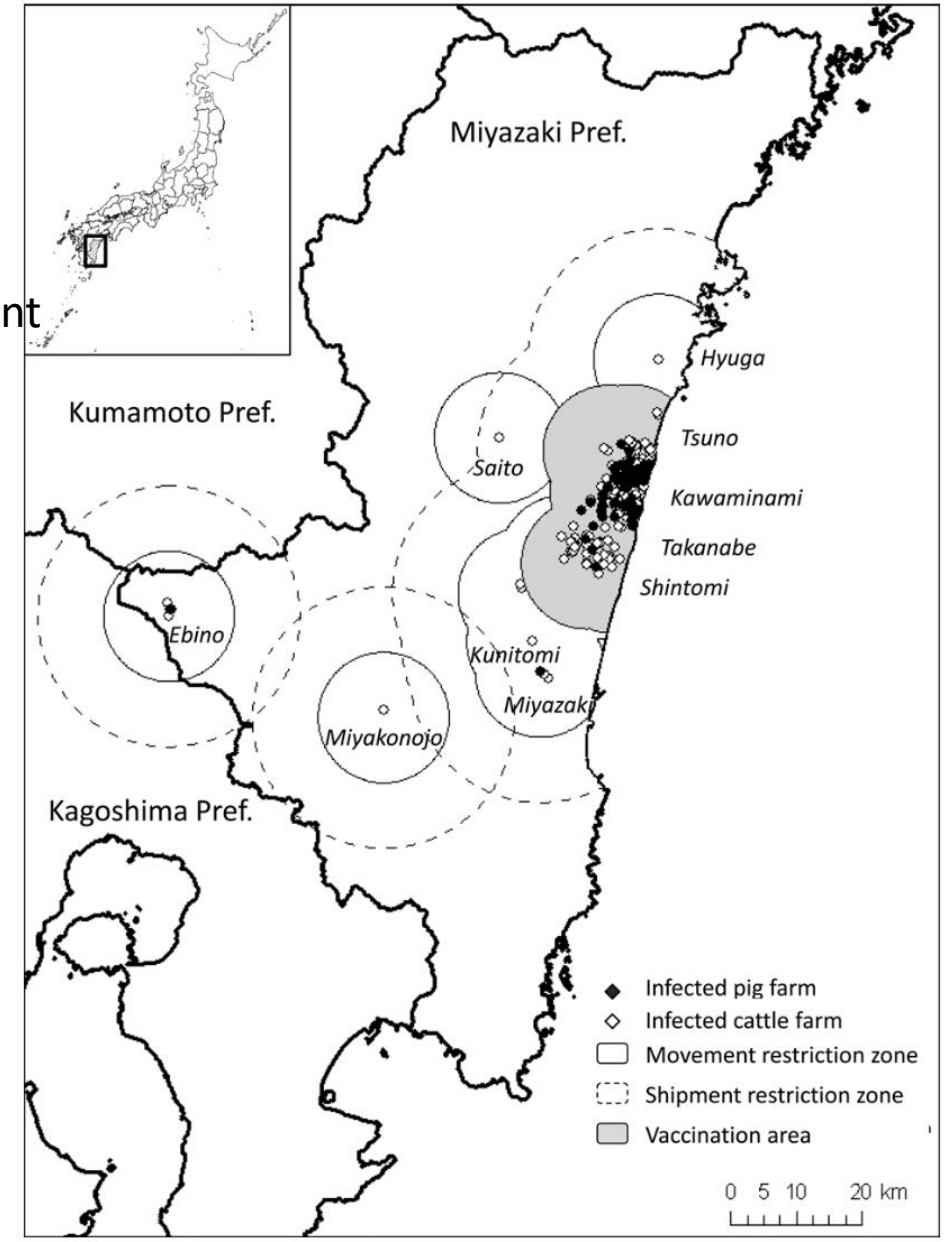


Fig. 3. Infected farms, movement restriction zones and vaccination zone during the 2010 FMD epidemic in Japan.

When pigs were vaccinated, they resisted infection with a standardized dose of FMDV, but when exposed to nonvaccinated pigs infected with FMDV (**high concentration of virus**) , vaccinated pigs became infected and transmitted the virus to other pigs in the same room.

→ Quantification of transmission

1. separation of pigs significantly reduced FMDV transmission
2. Vaccination can block transmission between pens, if a good-quality vaccine is used

Virus stability

Successful isolation of infectious FMDV from vesicular epithelium of pig carcasses stored at 4 °C for 11 weeks

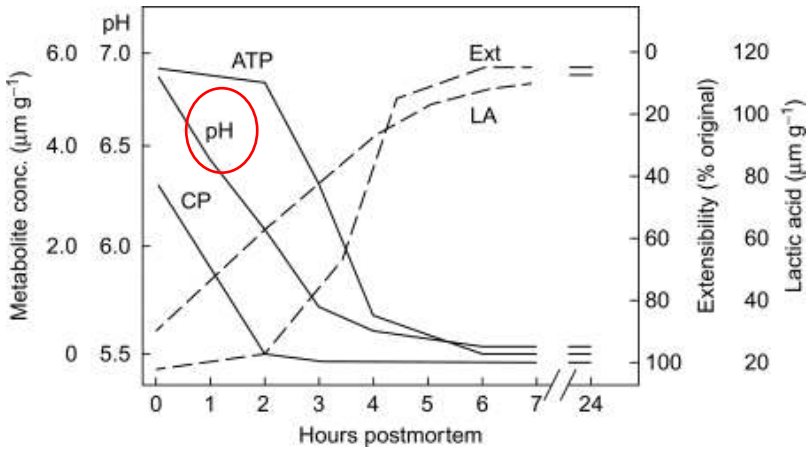
Most FMDV strains are **stable at lower temperatures** and at **pH 7–8**.

- FMDV remain infectious in the environment for weeks depending on the matrix, the initial concentration of the virus, and the ambient conditions (relative humidity, temperature, pH…)

FMDV are **resistant** to detergents and organic solvents.

- Alkaline, acidic and oxidizing disinfectants are effective.

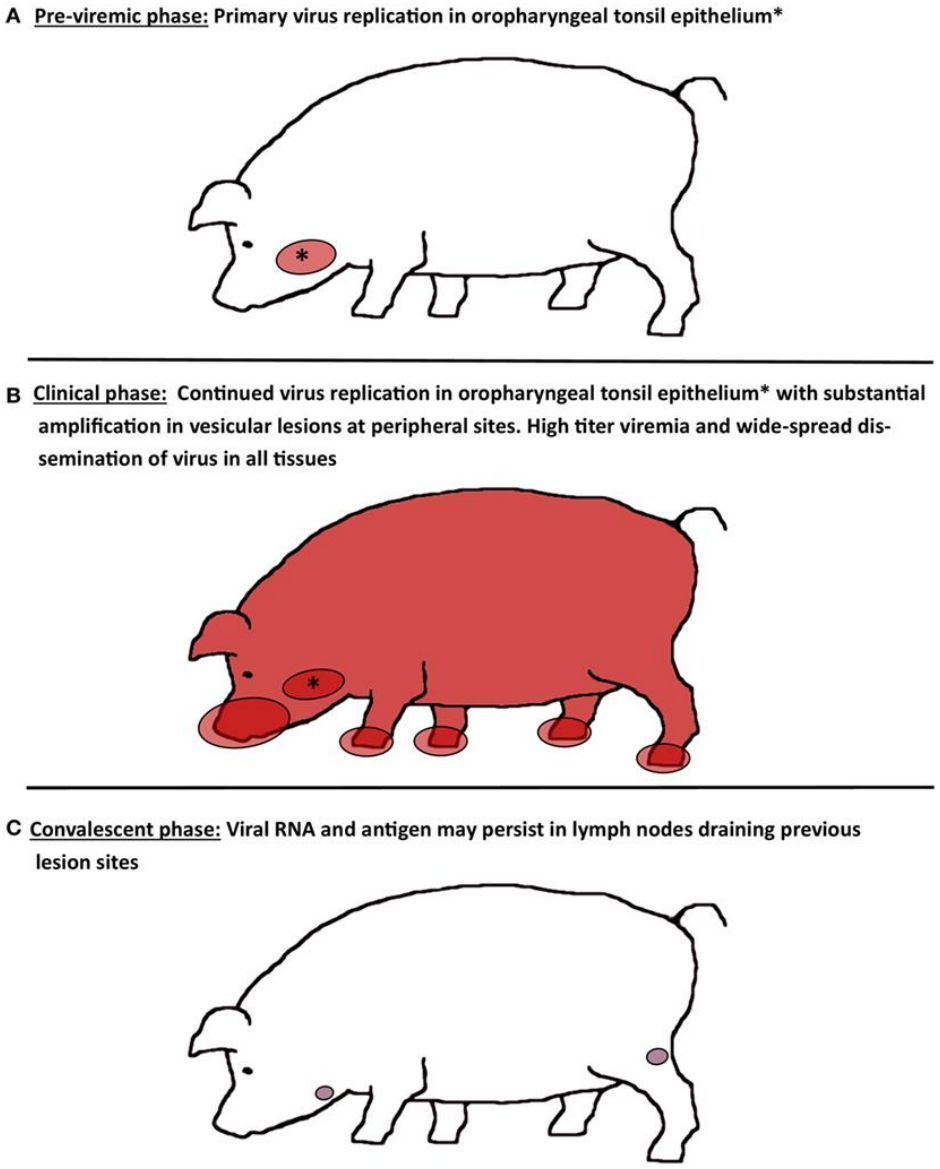
After death, the pH of muscles decreases, but that of bone marrow, lymph nodes or certain organs, does not.
→ Composting of pig carcasses!!



Reproduced from Greaser, M. L. (2001). Postmortem muscle chemistry. In: Hui, Y. H., Nip, W.-K., Rogers, R. W. and Young, O. A. (eds.) Meat science and applications, pp. 21–37. New York: Marcel Dekker.

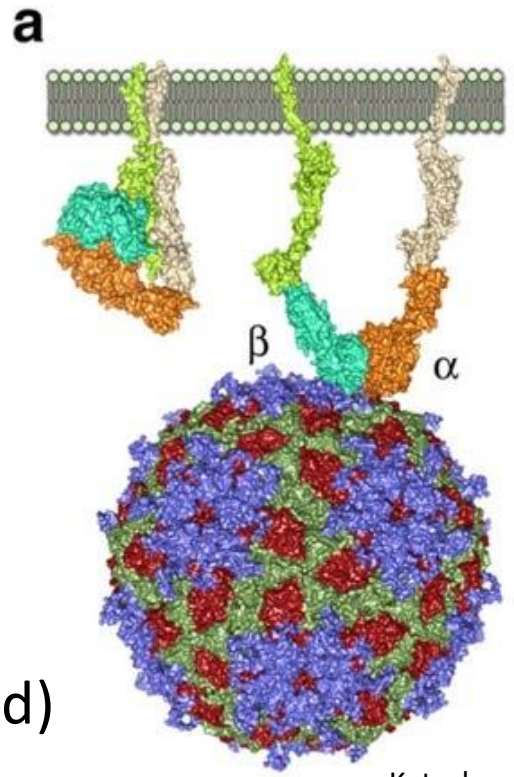
Pathogenesis

- **Varies according to host species**, dose, virus strain, farming condition…
- Shedding: secretion & vesicles
- Entry: Oro-^{pig} vs naso-^{ruminants}
  - -- Pharyngeal mucosa (multiply) → LN → viremia → seeding to 2nd sites (multiply)
  - -- Damaged integument



Stenfeldt et al., 2016

Pathogenesis



Kotecha et al., 2017

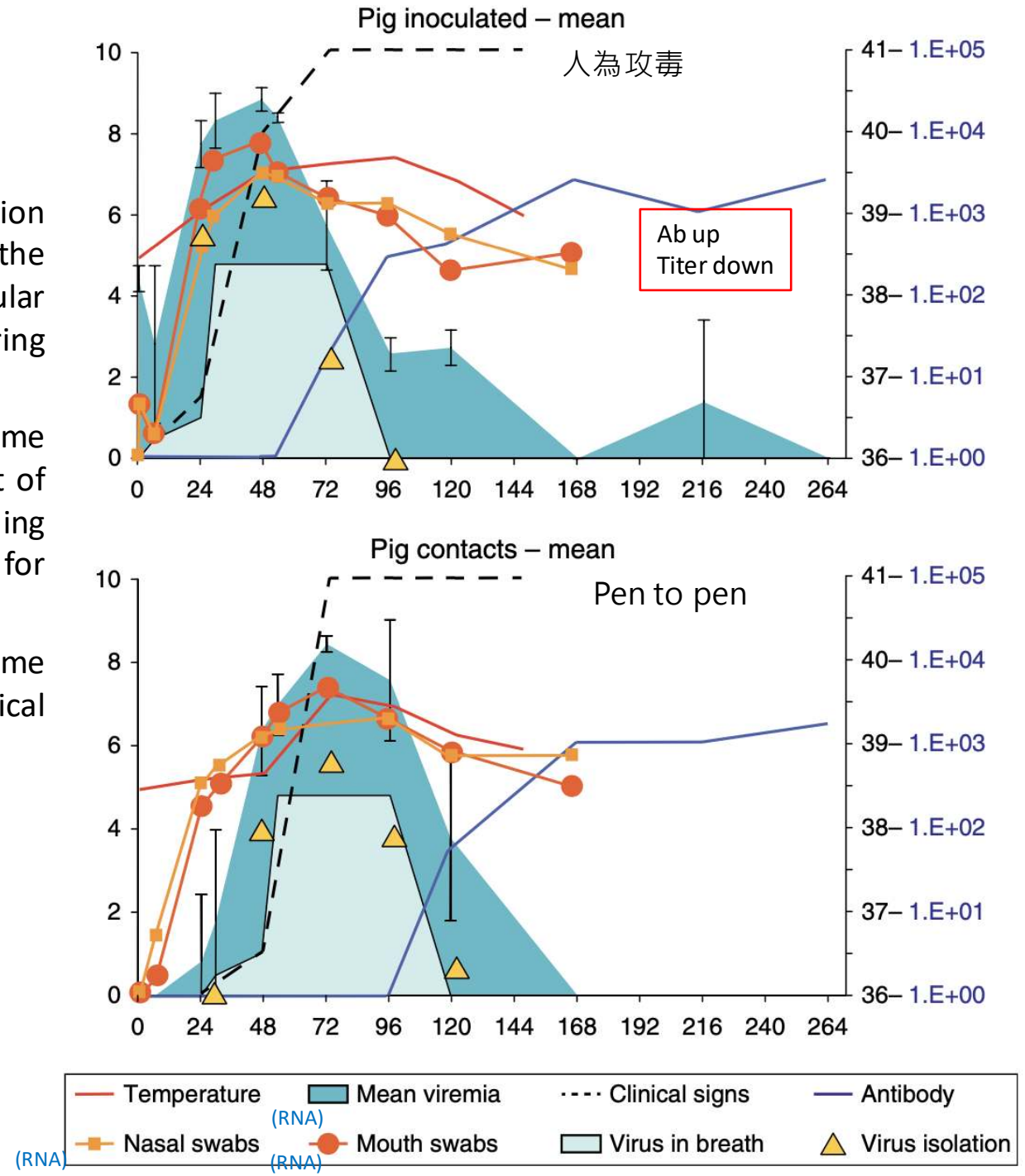
- Cellular receptor:
  - -- **Integrin**: mostly αvβ3 & **αvβ6** —– **VP1**
  - -- **Heparan sulfate** (HS) —– VP3 (TC-adapted)
  - -- J Jumonji C-domain Containing Protein 6 (JMJD6)

    —– mutated VP1 (TC-adapted)
  - -- Unknown receptor in MQ —– VP1

- Airborne viral excretion coincided with the appearance of vesicular lesions and occurred during the viremic phase
- **Latent period**: the time elapsed from the moment of infection to viral shedding reaching levels required for transmission
- **Incubation period**: the time from infection to first clinical signs of disease appear

FMDV: latent < incubation

沒症狀就在排毒



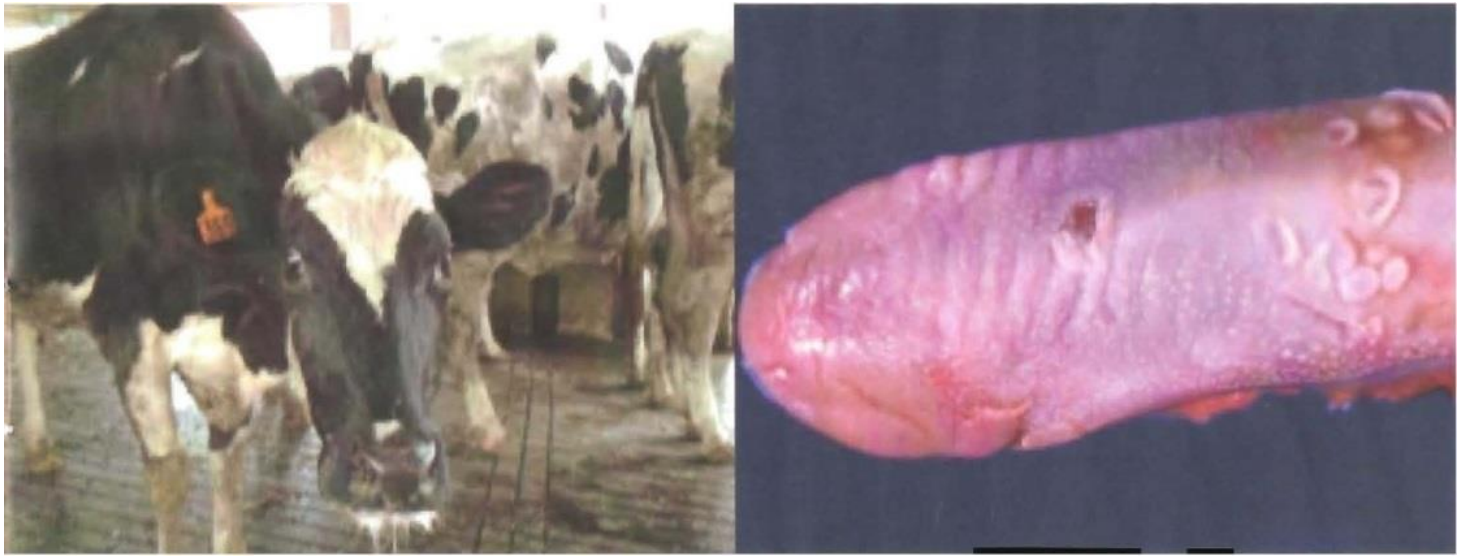
Clinical Presentation–Typical

- Incubation time:
  - -- Varies depending on host species, dose, virus strain, immunity…
  - -- ~14 days for farm-to-farm spread (airborne & indirect contact)
  - -- *How about direct contact?*
- **High morbidity** (up to 100%) but generally **low mortality**
- Clinical signs:
  - -- The severity varies depending on host species, age, virus strain…
  - -- Acute **fever**
  - -- **Blisters** on the nose, tongue or lips, inside the oral cavity, between the toes, above the hooves, on the teats and at pressure points on the skin.

Depression, drooling, chomping, lameness, unwilling to walking

Clinical Signs





Clinical Signs



https://www.youtube.com/watch?v=Znygl3V4pdI

Lesions

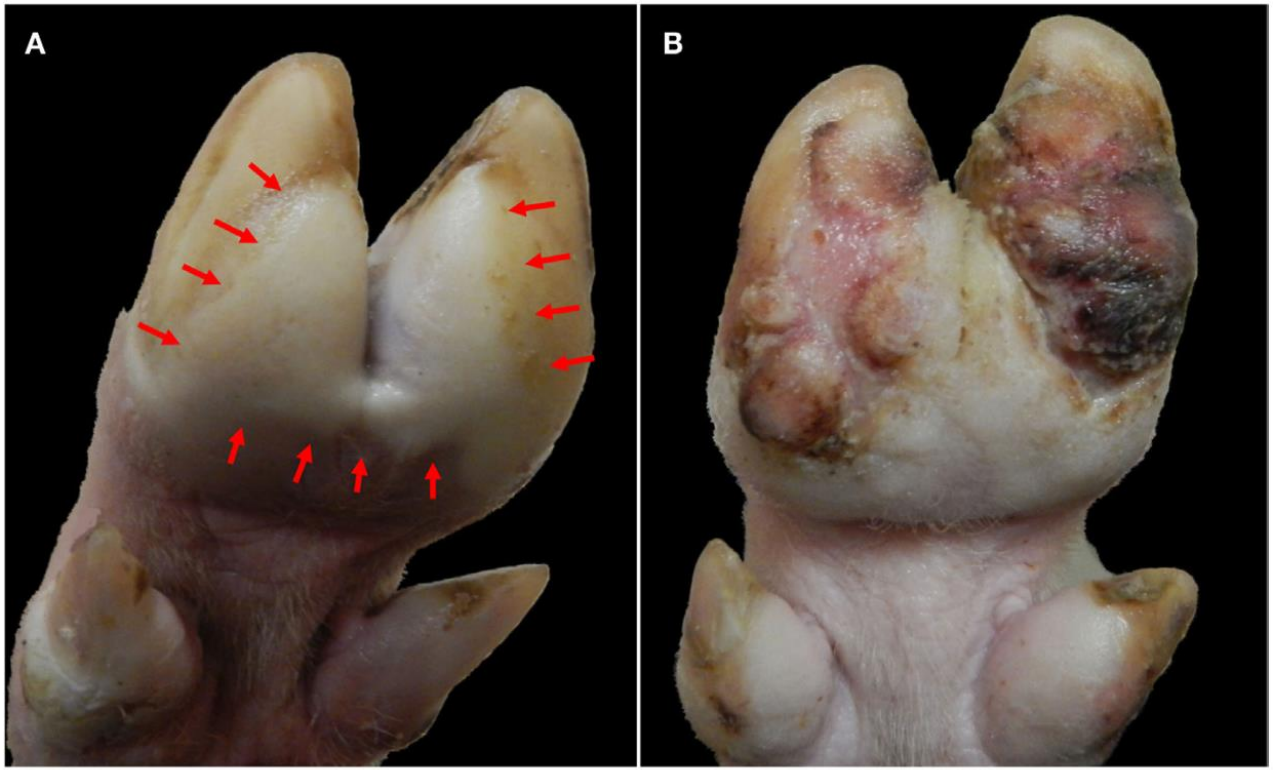
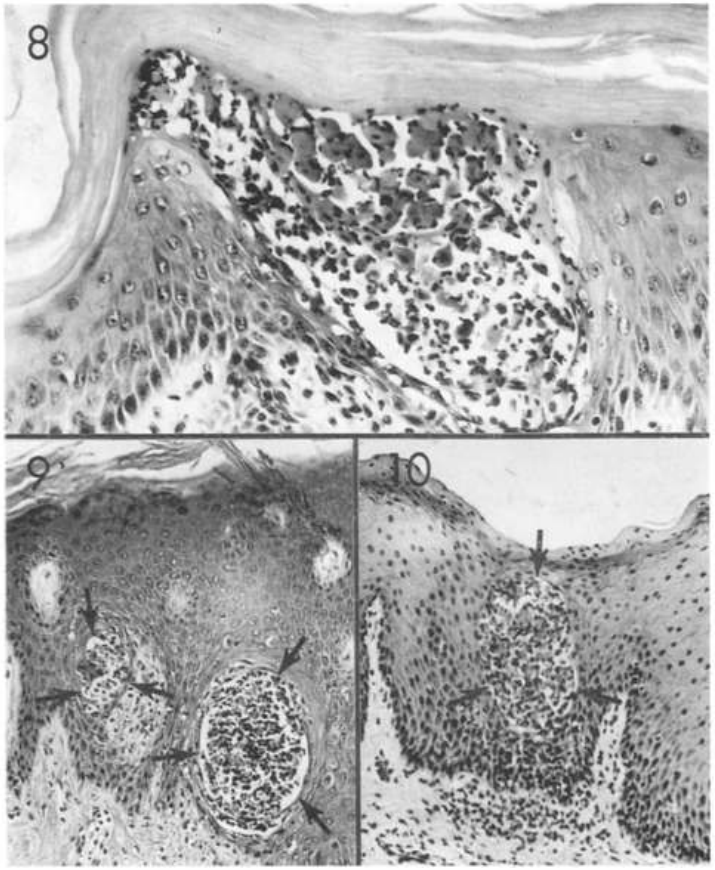


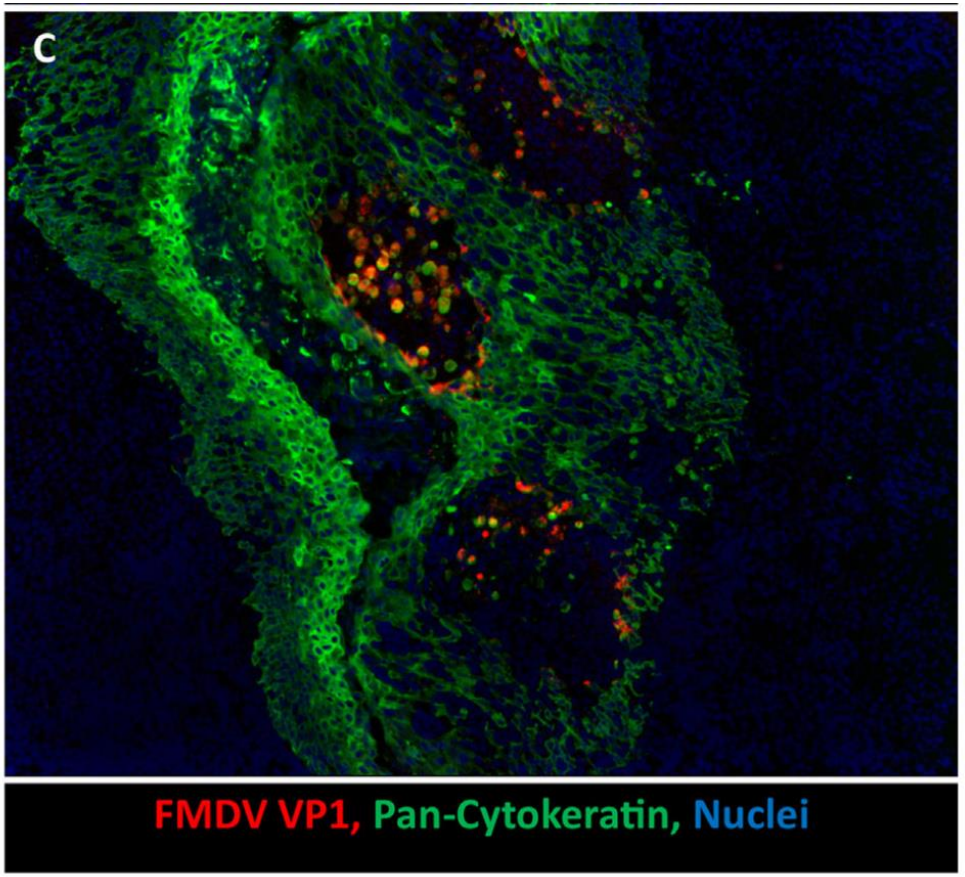
FIGURE 3 | Progression of foot lesions in pig infected with FMDV A_{24} Cruzeiro at 2 (A) and 24 (B) days after intraoropharyngeal inoculation. **(A)** Vesicular lesion on solar aspect of hind foot at 2 dpi. Blanching (white) epithelium (delineated within arrows) extends across both heel bulbs and interdigital skin with clear demarcation from normal skin. **(B)** The same animal at 24 dpi. Proliferative dyskeratotic scar tissue has replaced sloughed epithelium.

Stenfeldt et al., 2016

Lesions



Gailiunas 1968



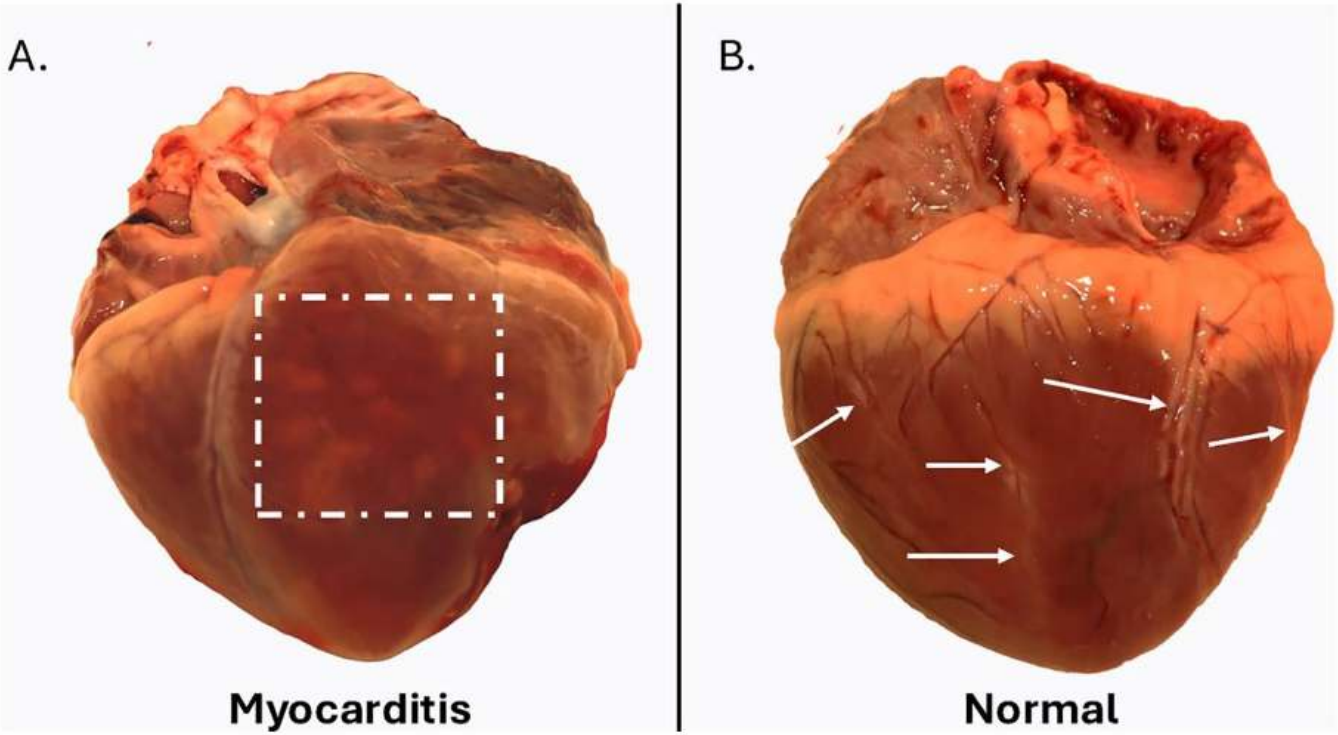
Stenfeldt et al., 2016

- Ballooning degeneration and increased eosinophilia of the keratinocytes in the stratum spinosum.
- Necrosis of affected keratinocyte followed by infiltration of mononuclear cells & granulocytes.
- Formation of vesicles.

Clinical Presentation–Atypical

-- Sudden death in piglets due to myocarditis (**Tiger heart**)

-- Abortion

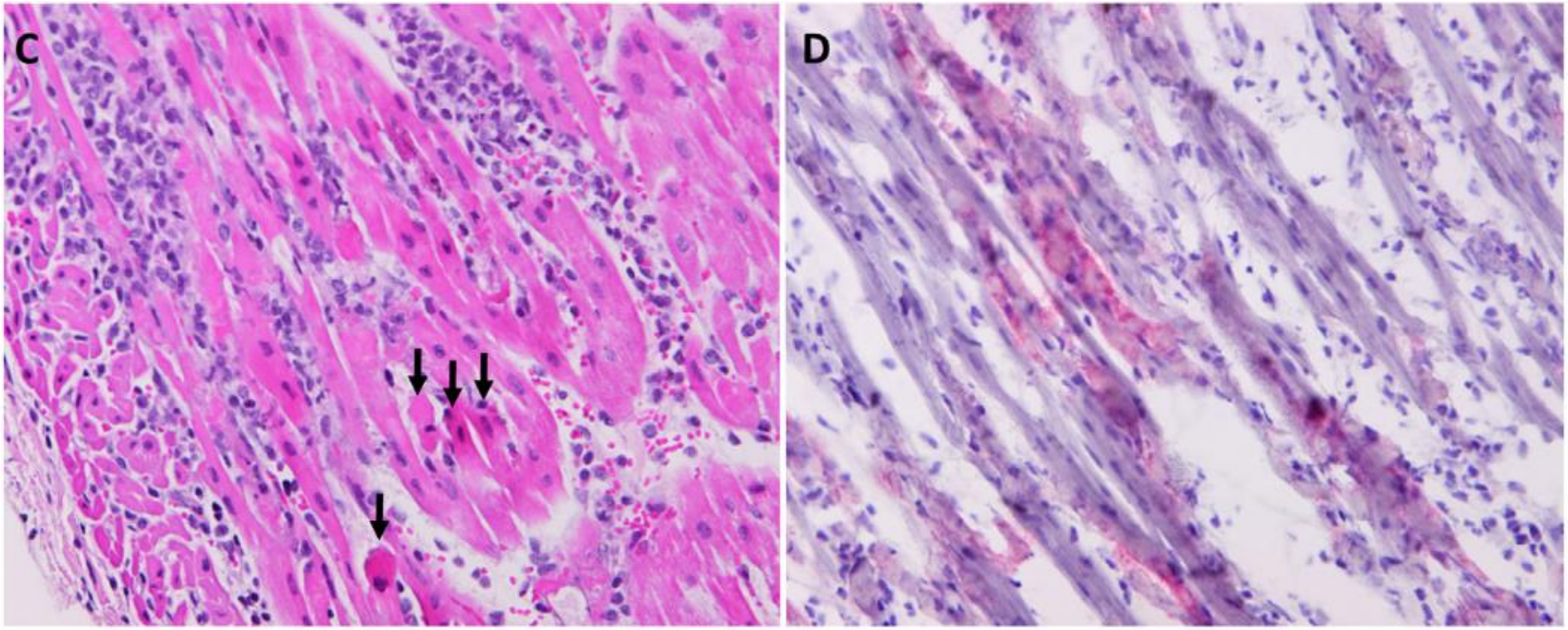


Gross image of porcine heart with confirmed FMDV-associated myocarditis. Note the multifocal pallor on the surface of both right and left ventricles.

Relatively **uncommon** lesions. (但國考愛考)

Stenfeldt et al., 2025

FMDV-associated Myocarditis



(C) Right ventricle of pig found dead at 5 days post infection with FMDV A24 Cruzeiro. Interstitial edema and mixed mononuclear infiltrates consisting of lymphocytes, large macrophage-like cells, and scarce neutrophils. Myocyte necrosis and fragmentation (arrows). (D) Immunohistochemical staining demonstrating localization of FMDV within cardiomyocytes. Anti-FMDV capsid monoclonal antibody (red).

Stenfeldt et al., 2016

Clinical Presentation–Subclinical

- Neoteric SI

A substantial spread of FMDV may occur

  -- Early (acute) phase FMDV infection **without clinical signs** of disease

  -- Viral **shedding** in oronasal secretions → transmissible

  -- Pre-existing immunity from **vaccination**, **previous exposures**, or natural **resistance** to clinical FMD

- Persistent SI (mainly in ruminants but not suids)

  -- Carrier state, occurs after either clinical or subclinical primary infection, regardless of vaccination

  -- Low-level viral replication → **minimal to no viral shedding**

  -- Contagion associated with FMDV carriers is believed to be minimal

  -- May contribute to recombination and diversity of FMDV

Differential Diagnosis

	Vesicular stomatitis 水疱性口炎 zoonosis	Vesicular Exanthema of Swine 水疱疹	Swine Vesicular Disease 豬水疱病	Seneca Valley Virus 矽尼谷病毒感染
Virus	Rhabdoviridae	Caliciviridae	Picornaviridae	Picornaviridae
Natural host	Horse, cattle, swine	Swine	Swine	Swine
Endemic region	America	Eradicated	Europe and Asia (including Taiwan)	Globally, including Taiwan

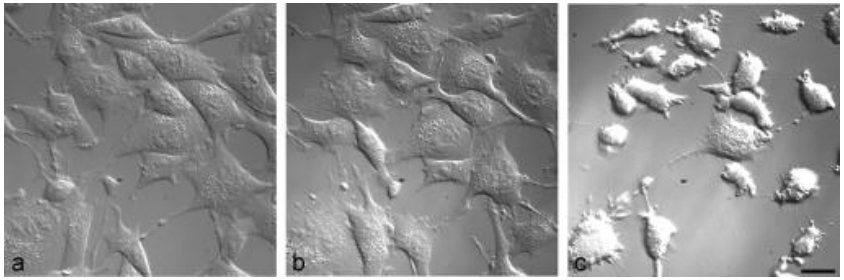
- The vesicular lesions are **indistinguishable** from those induced by an FMDV infection.
- Vesicular diseases in pigs **must be assumed to be FMD** until proven otherwise.

Laboratory Diagnosis

Biosafety level 3

Clinical signs and lesions of FMD are **not** diagnostic since they look similar in all swine vesicular viral diseases.

- Virus detection (**vesicular materials**):
  - -- Identification of (infectious) virion: EM, virus isolation
    - **Susceptible cell lines such as BHK-21, LFBK-αVβ6 or IB-RS-2
  - -- Identification of antigen: ELISA (confirmation + serotyping)
  - -- Identification of nucleic acids: RT-PCR, RT-qPCR



Armer et al., 2008

Laboratory Diagnosis

Biosafety level 3

- Serological test:

  Purposes

  1. to certify individual animals prior to import or export (i.e. for trade)
  2. to confirm suspected cases of FMD
  3. to estimate the prevalence of infection or to substantiate its absence
  4. to demonstrate the efficacy of vaccination

  -- Virus neutralization

  -- ELISA for Ab detection

  -- Detect Ab against non-structural protein (NSP)

  → **d**ifferentiate **i**nfected from **v**accinated **a**nimals (**DIVA**)

DIVA

Whole package:

DIVA vaccine + DIVA test

- Selectable marker (Ag)

  -- In natural infection:

  Good antigenicity and longevity

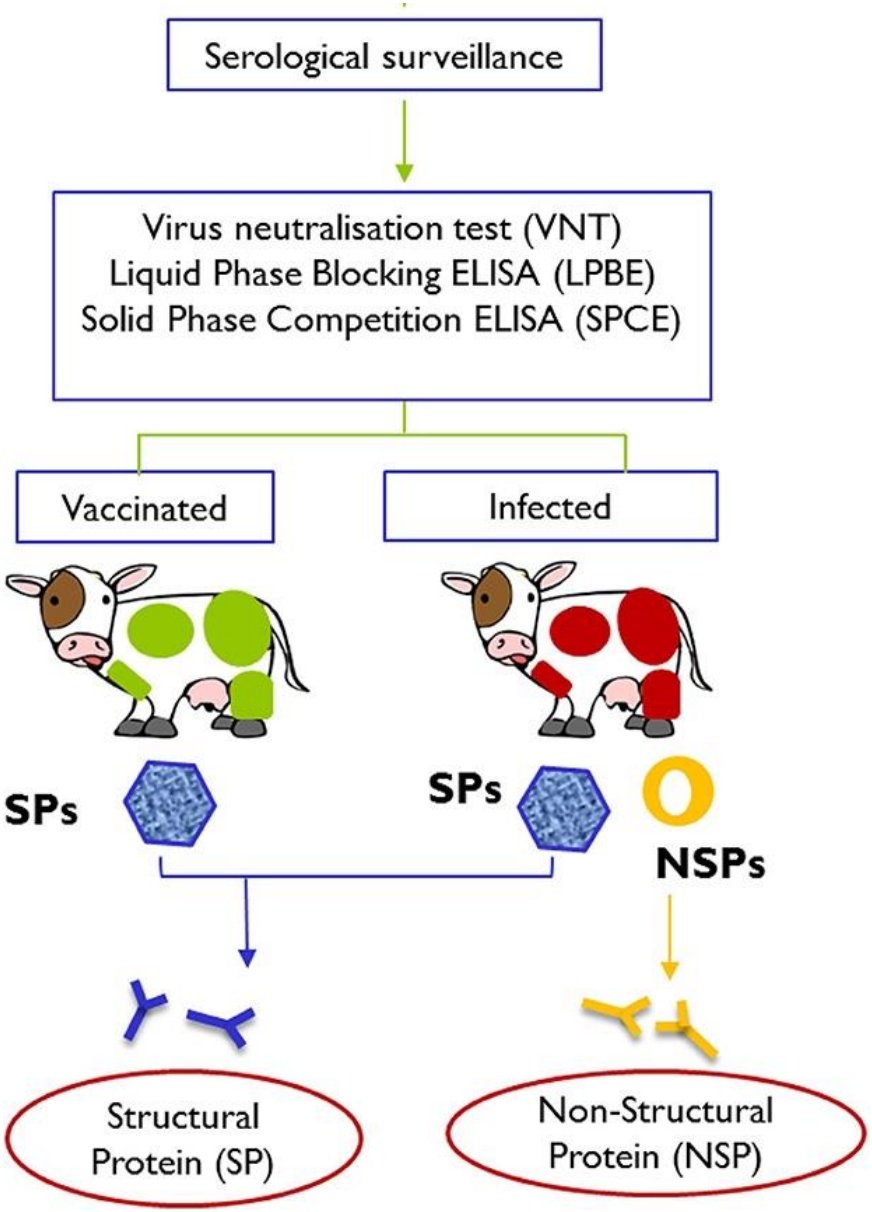
  -- Upon vaccination

  Absence of immunoreactivity

- ex. 3A3B NSP for FMD DIVA vaccine

  NS3 for CSF DIVA vaccine

  gE for PR & IBR DIVA vaccine



Wong et al., 2016

Prevention & Control

The most common spreading routes of FMDV:

1. movement of infected animals (direct)
2. feeding of contaminated animal products to the herds
3. movement of virus on fomites or vectors (indirect)

→ Stamping out of infected & suspect animals

→ Movement restrictions & biosecurity procedures

→ ± Vaccination (mass coverage vs specific subpopulation)

→ Continuous surveillance

114 年度口蹄疫擴大監測結果統計表

統計資料更新時間：114 年 9 月 17 日

動物種類	已檢驗場數	檢驗結果	備註
豬	215 場	陰性 215 場	
		陽性 0 場	
草食動物	194 場	陰性 194 場	
		陽性 0 場	

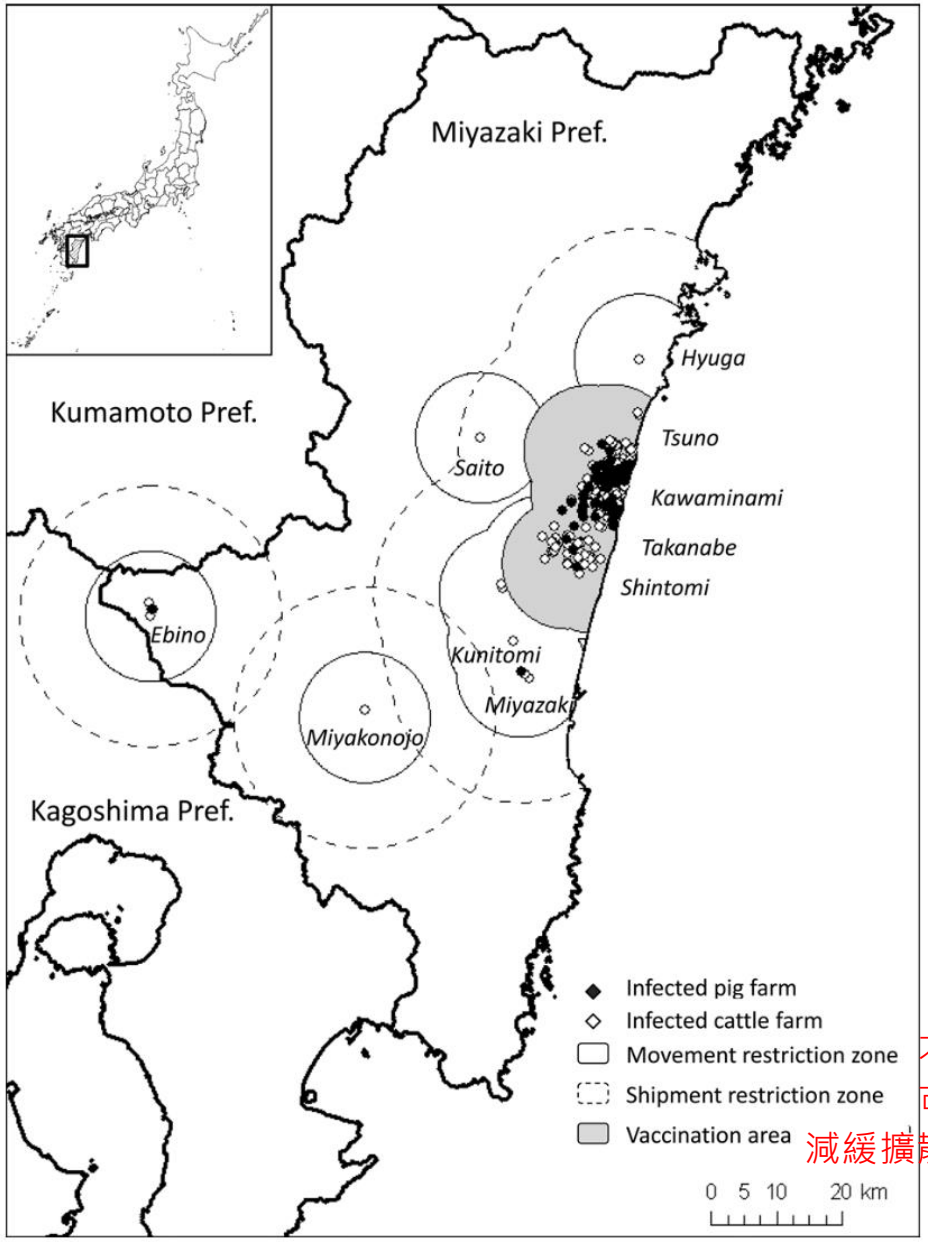
監測結果顯示國內畜牧場 114 年度迄今無發生口蹄疫案例。

https://www.aphia.gov.tw/ws.php?id=22285

Vaccination

- Current facts:
  -- Most commercial vaccines for FMD are **inactivated vaccines** which **do not prevent primary infection**.
  -- Vaccination reduce viral shedding & thus slow down spreading
  -- Vaccination strategy is mainly applied for **emergency use** and must combine with **strict biosecurity**.

*Outbreak in Japan 2010



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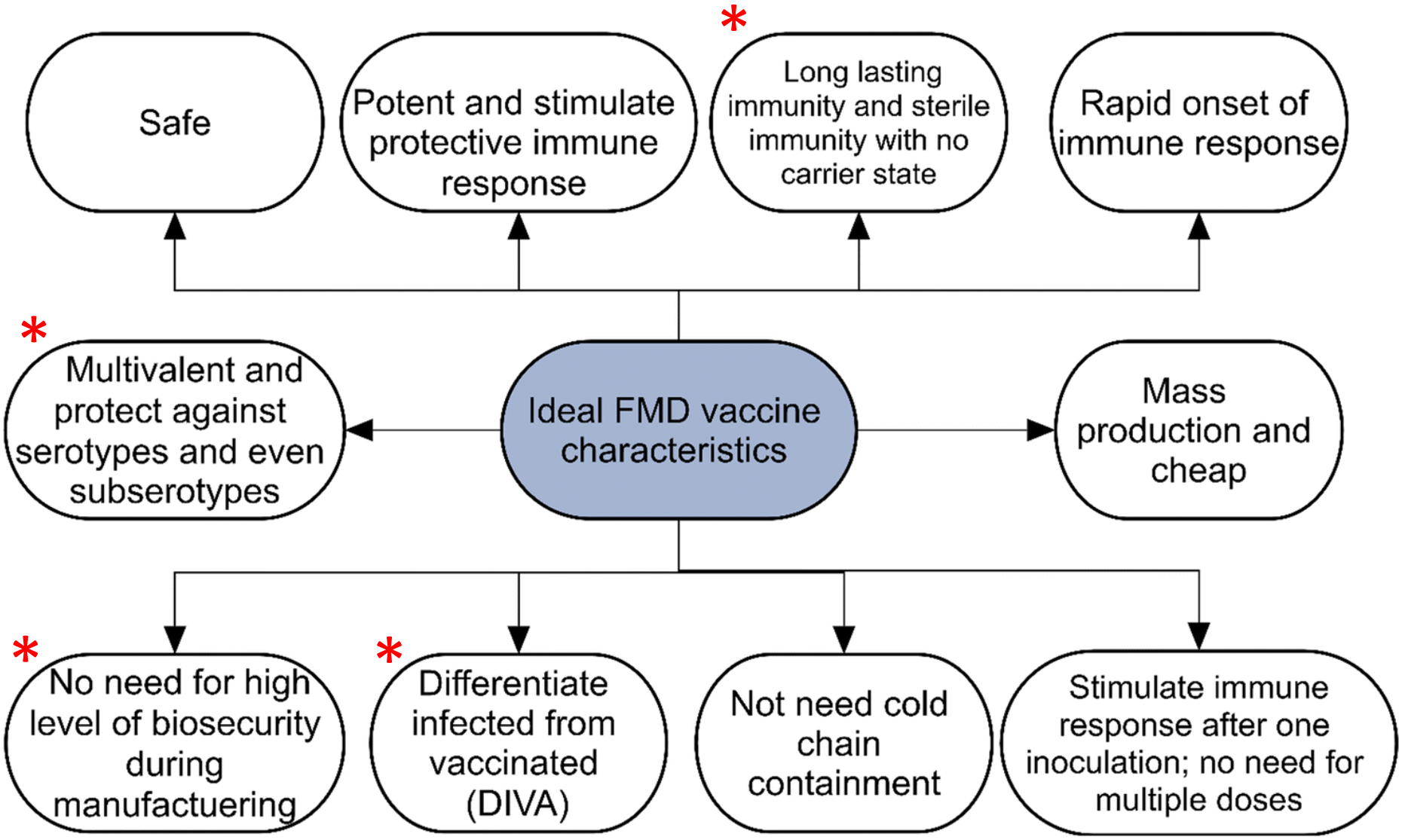
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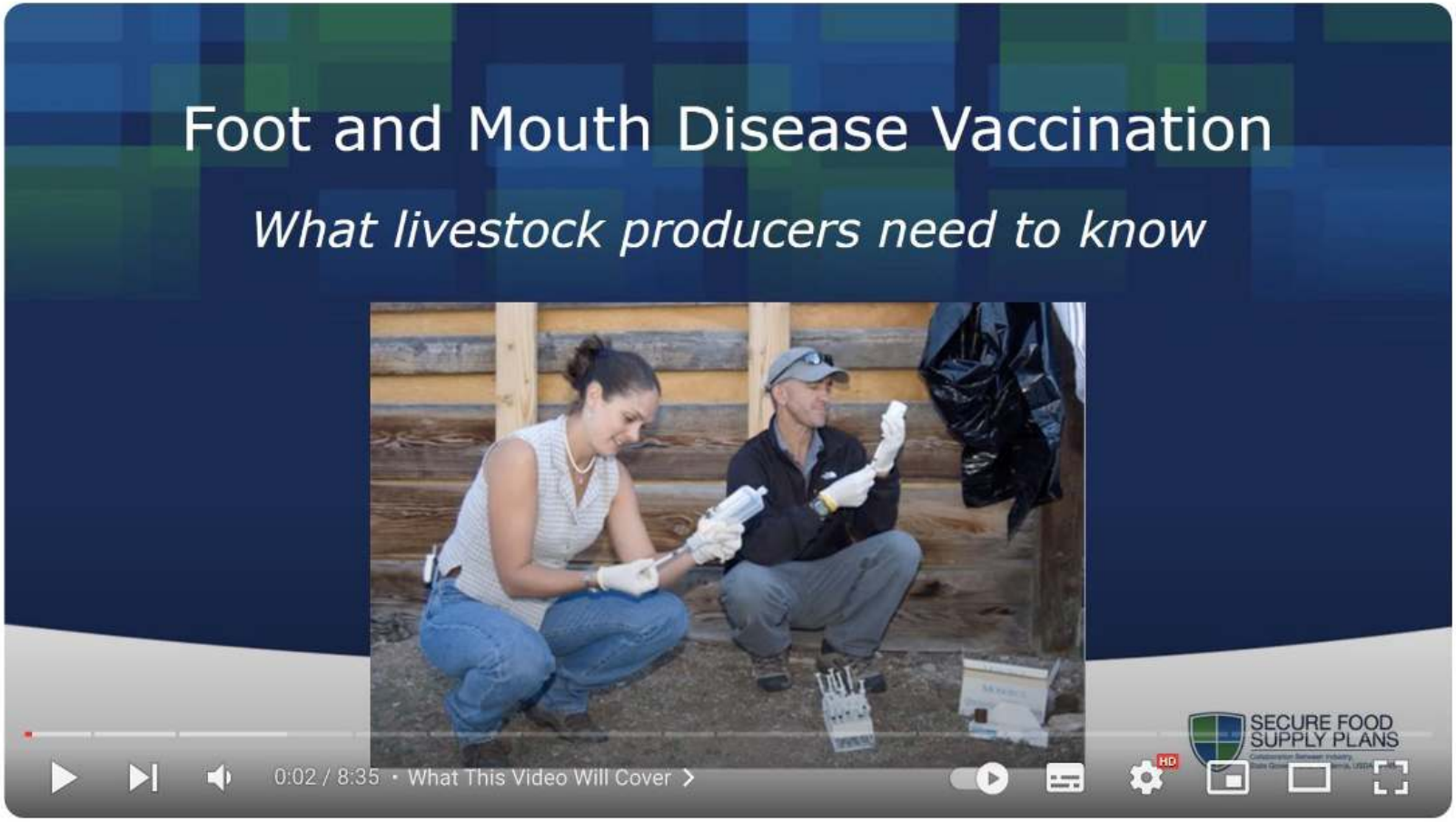
減緩擴散，最終還是被撲殺

Fig. 3. Infected farms, movement restriction zones and vaccination zone during the 2010 FMD epidemic in Japan.

Vaccination

- Consideration:
  - -- Seven serotypes with **no or limited cross-protection**
  - -- Spread easily
  - -- Protective response last only around 4-6 m
  - -- Interference from maternal immunity before 8-12 weeks of age
  - -- Vaccinated animals may develop **subclinical infection**


Safe
Potent and stimulate protective immune response
*
Long lasting immunity and sterile immunity with no carrier state
Rapid onset of immune response
*
Multivalent and protect against serotypes and even subserotypes
Ideal FMD vaccine characteristics
Mass production and cheap
*
No need for high level of biosecurity during manufactuering
*
Differentiate infected from vaccinated (DIVA)
Not need cold chain containment
Stimulate immune response after one inoculation; no need for multiple doses



Foot and Mouth Disease Vaccination Overview

https://www.youtube.com/watch?v=MKf-aMgb-y0

FMD Outbreak in the UK 2001



Hoof and Mouth Outbreak & Impact, by Dr. Chris Morley

https://www.youtube.com/watch?v=6SN917cUc5A

Comparative Immunology, Microbiology and Infectious Diseases
Volume 25, Issues 5–6, October 2002, Pages 331-343

The foot and mouth disease (FMD) epidemic in the United Kingdom 2001

Gareth Davies

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Abstract

The foot and mouth disease epidemic commenced in February 2001 when diseased pigs were identified in an abattoir. The infection had become widespread in sheep in England and Wales before this discovery. It was decided to eradicate the disease by slaughter rather than use vaccine. The virus was a Pan-Asia O strain that caused few lesions in sheep and this made the identification of infected flocks very difficult leading to a long drawn-out epidemic. Over four million animals were slaughtered in 2000 herds and flocks. The last outbreak was in September.

The consideration of vaccination against FMD outbreak in the UK 2001

Comparative Immunology, Microbiology and Infectious Diseases
Volume 25, Issues 5–6, October 2002, Pages 331-343

The foot and mouth disease (FMD) epidemic in the United Kingdom 2001

Gareth Davies

There are two alternative strategies in eradicating FMD: slaughter and vaccination. Vaccination has never been employed in the UK although preparations were made to distribute vaccine in 1967. In this present epidemic the issue of vaccination arose a number of times. The first time was during March when the full extent of the epidemic became clear. The veterinary service made preparations for a vaccination campaign but Government ministers said they were reluctant to resort to vaccination.

The reasons for this decision, as given by Ministers, were that vaccination did not have the support of the majority of the farming community, veterinarians, the wider food industry and even consumers.

The Government had established a committee of scientists to provide scientific advice (the FMD science group), under the chairmanship of the government's Chief Scientific Advisor and they were also reluctant to propose vaccination, partly on the ground **that there was bound to be a delay in immunising the population** at risk and partly because **they believed that vaccination would give rise to 'carriers' that would cause further outbreaks and prolong the epidemic**. This 'canard' was eventually corrected but by that time the main opportunity to curtail the epidemic by vaccination had been lost.

The number of outbreaks reported each day continued to increase throughout March and the total reached 1000 by the end of that month. Permission for a limited vaccination programme in the worst affected areas: Cumbria and Devon had been granted by the European Commission (Decision 2001/257/EC) and the FMD science group eventually recommended vaccination of cattle in these areas prior to the animals leaving their winter quarters, when they might encounter infected sheep. Some 500,000 doses of high potency vaccine were prepared at the Institute for Animal Health Pirbright and distributed. In the event this vaccine was not used, largely because of **continuing resistance from within the livestock and food industries**.

The farming community was divided on the issue**. The larger farmers, particularly sheep farmers who depended on an export trade**, felt that vaccination would cause **long term damage**. Many of the **smaller farmers and particularly producers of organic meat and milk products feared that the slaughter strategy would destroy the herds** that they had laboriously created over the last decades and strongly supported vaccination.

2.下列何種豬水疱性傳染病會感染人？

(A)Seneca Valley virus infection

(B)swine vesicular disease

(C)vesicular stomatitis

(D)vesicular exanthema of swine

在使用口蹄疫疫苗地區,可針對下列那一ELA抗體進行監測,以判定是否有野外病毒之感染?

(A)結構性蛋白VP1
(B)結構性蛋白VP2
(C)結構性蛋白VP4
(D)非結構性蛋白

口蹄疫病毒O TAIWAN 99會感染下列何種動物？

(A) 僅感染牛及羊
(B) 僅感染羊及豬
(C) 僅感染豬
(D) 牛、羊及豬均會感染

關於口蹄疫之敘迹,下列何者正確?①感染偶蹄動物,例如:牛,羊,豬②此病毒為單一血清型③為傳染力低的病原④人類對此病毒感受性不高,但扮演此病毒攜帶者之角色

(A) ①②
(B) ①④
(C) ②③
(D) ③④

有關水疱性疾病(如:口蹄疫、水疱性口炎、水疱疹與豬水疱病)，何者正確?

(A) 四種的水疱性疾病，由病理變化無法區別
(B) 口蹄疫可感染牛、羊、豬及馬
(C) 水疱疹可感染牛及馬
(D) 豬水疱病與水疱性口炎同屬一個病毒群

下列有關口蹄疫之敘述，何者最正確？

(A)牛為口蹄疫病毒最主要增幅者
(B)經呼吸道感染，牛比豬需更高的最低感染劑量才能有效感染
(C)在一般情形下口蹄疫病毒可進行長距離空氣傳播
(D)口蹄疫病毒感染每隻每天經由呼吸道排出之病毒量，豬較牛多

口蹄疫病毒有7個血清型，下列何者非口蹄疫病毒之血清型？

(A) O型
(B) A型
(C) Asia 1型
(D) D型

口蹄疫病毒有7個血清型，下列何者非口蹄疫病毒之血清型？

(A) O型
(B) A型
(C) Asia 1型
(D) D型

下列何動物在感染口蹄疫病毒後之平均帶毒時間最長？

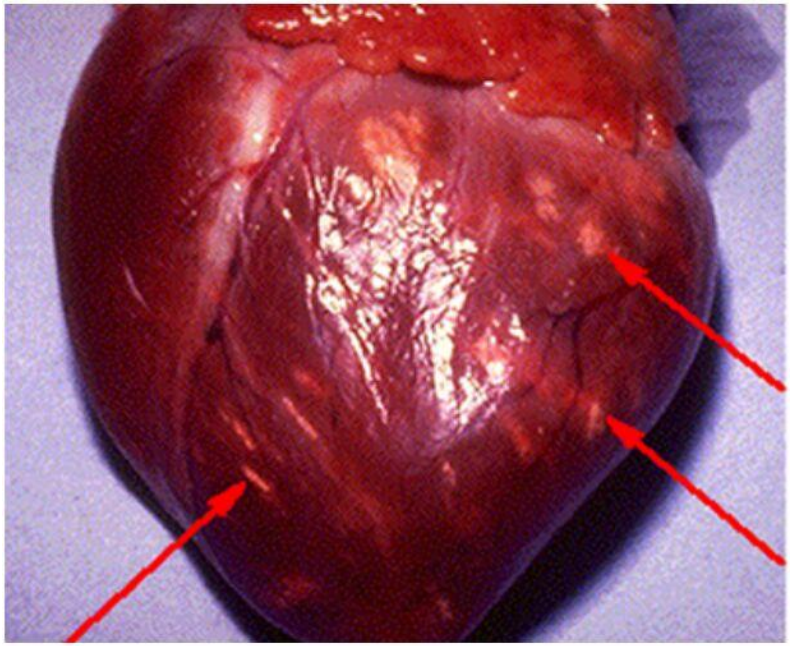
(A) 豬
(B) 牛
(C) 綿羊
(D) 山羊

口蹄疫病毒O TAIWAN 97會感染下列何種動物？

(A) 僅感染牛及羊
(B) 僅感染羊及豬
(C) 僅感染豬
(D) 牛、羊及豬均會感染

仔豬口蹄疫引起下圖心臟病變，稱之為：

(A) white muscle disease
(B) cor villosum
(C) shaggy heart
(D) tiger heart



Swine Vesicular Disease (SVD)

Introduction

- First reported in Italy in 1966
- A **vesicular** disease of swine that is clinically indistinguishable from foot-and-mouth disease.
- The causative agent, swine vesicular disease virus (SVDV) is suspected be evolved from human enterovirus.
- SVD has been **removed** from the WOAH listed diseases.
- “In the FMD outbreak in 1997 in Taiwan, the presence of SVDV in the country probably delayed the reporting of FMD cases, which may have led to more FMDV-infected farms.”---Diseases of swine

Etiology

Swine vesicular disease virus (SVDV)

- Genus *enterovirus*, family of *Picornaviridae*
- **Non-enveloped** (1A-D), icosahedral, 30-32 nm
- ss, positive-sense RNA genome, 7.4 kb in length
- **A single serotype** with at least four antigenic variants
- Swine are the only natural host for SVDV.
- Not zoonotic

Transmission

latent < incubation

Affected pigs may excrete virus up to 48 hours **before** the onset of clinical signs.

SVDV is exceptionally **stable** outside the hosts.

- Movement of sub-clinically infected pigs
  - -- direct contact with infected animals
  - -- contact with a contaminated environment (ex drainage system)
  - -- Pen disease
- Ingestion of contaminated food (swill feeding)

Virus Stability

- SVDV is stable over a **wide range of pH**.
- SVDV is also resistant to detergents and organic solvents.
- SVDV remains infectious for months in carcasses and processed meat.
- Only sodium hydroxide (1%) was able to inactivate SVDV adequately.

Pathogenesis

- SVDV enters through the **skin** or the **mucous membrane** of the digestive tract
- SVDV replicate primarily in **epithelial cells**.
- Lesions are mainly observed in epithelial tissues and is usually mild or subclinical.
- Disease severity generally correlates with **age** and the environmental condition (ex concrete floor in humid conditions).
- Virulence factors in different strains are unknown.

TABLE I. SVDV Titre (−logTCID$_{50}$/ml) of Nervous Tissues and Plasma from Inoculated Microtitres Plates at One to Seven Days Postinoculation

Site	Postinoculation Days 1	2	3	4	5	6	7
Olfactory	2.33	5.7	6.0	4.0	0	ND	0
Prefrontal	0[a]	0	4.8	3.5	0	ND	0
Paracentral	0	0	3.0	0	0	ND	0
Occipital	2.7	0	3.2	0	0	ND	0
Corpus callusum	2.7	ND	0	0	0	ND	0
Hypophysis	0	ND	0	ND	0	ND	0
Cerebellum I[b]	0	0	0	ND	1.8	ND	0
Cerebellum II[c]	0	ND	0	ND	0	ND	0
Cerebellum III[d]	1.5	0	2.5	ND	1.7	ND	0
Cerebellum IV[e]	0	0	2.7	ND	0	ND	0
Thalamus	0	3.2	3.8	0	0	ND	0
Peduncle	2.4	3.2	3.8	3.2	0	ND	0
Pons	2.3	3.5	3.6	0	0	ND	0
Medulla oblongata	2.5	3.8	3.6	0	0	ND	0
Spinal cord	2.7	3.8	4.3	0	0	ND	0
Plasma	4.5	4.7	2.5	0	0	ND	0

ND = not determined
[a]Values less than 1.33
[b]Anterior portion
[c]Antero-medial portion
[d]Postero-medial portion
[e]Posterior portion

TABLE II. SVDV Titre (−log TCID$_{50}$/ml) of Epithelial Tissues, Lymphoreticular Tissues, Muscles and Plasma from Inoculated Microtitre plates at One to Seven Days Postinoculation

Site	Postinoculation Days 1	2	3	4	5	6	7
Snout	6.0	6.5	7.2	7.0	5.9	4.5	3.8
Lip	6.8	6.5	6.4	5.0	5.6	4.0	5.0
Tongue	7.2	6.8	6.4	6.5	6.7	6 4	6.5
Coronary band	5.4	6.0	7.2	5.0	5.3	4.0	3.3
Skeletal	4.0	5.4	5.0	4.0	2.7	ND	0
Cardiac	6.9	5.8	6.4	4.5	3.3	ND	0
Tonsil	5.7	6.0	5.7	5.3	4.7	4.5	3.5
Thymus	6.1	4.4	3.2	0[a]	0	ND	0
Spleen	4.6	4.2	2.3	ND	1.7	ND	0
Suprapharyngeal lymph node	4.1	4.4	4.3	ND	4.7	ND	0
Posterior mediastinal lymph node	4.9	4.7	3.9	ND	0	ND	ND
Bronchial lymph node	5.7	5.3	3.1	1.7	3.7	2.2	0
Mesenteric lymph node	4.5	ND	3.8	ND	0	ND	0
Plasma	4.5	4.7	2.5	0	0	0	0

ND = not determined
[a]Value less than 1.33

Clinical Presentation

- Short incubation time
- **Varying morbidity** and **low mortality**
- Clinical signs are similar to but **much milder** than those seen from the FMDV infection and may include:
  - -- Sudden appearance of lameness
  - -- Transient fever
  - -- Vesicles on the coronary band, typically at the junction with the heel, and interdigital spaces of the feet.
  - -- Vesicles around the snout (less common)
  - -- ~~Sudden death in piglets and abortion in sow~~

Not features of SVD

Lesions



Vesicular lesion on the nose.



Vesicular lesion on the coronary band.



Vesicular lesion in the interdigital space.



Vesicular lesion.

https://www.cabidigitallibrary.org/doi/full/10.1079/cabicompendium.62450

Differential Diagnosis

	Vesicular stomatitis	Vesicular Exanthema of Swine	Foot-and-mouth disease	Seneca Valley Virus
Virus	Rhabdoviridae	Caliciviridae	Picornaviridae	Picornaviridae
Natural host	Horse, cattle, swine	Swine	cloven-hoofed animals	Swine
Endemic region	America	Eradicated	Africa and Asia	Globally, including Taiwan

- Vesicular disease in pigs **must be assumed to be FMD** until proven otherwise.

Laboratory Diagnosis

- Virus detection (**vesicular materials**):
  -- Virion: EM, virus isolation
  -- Antigen: ELISA
  -- Nucleic acids: RT-PCR, RT-qPCR (for fecal sample in subclinical cases)
- Serological test:
  -- Viral neutralization
  -- ELISA for Ab detection

Prevention & Control

Currently **no** commercial vaccines available against SVD

- Proper preventative measure, ex strict hygiene, no swill feeding
- Routine surveillance
- Culling all infected and in-contact pigs

有關水泡性疾病，何者正確？

(A) 口蹄疫病毒主要感染偶蹄類動物，但可能出現僅感染豬之型別，如O/Taiwan/97
(B) 水泡性口炎（vesicular stomatitis），只會發生於豬及反芻動物，不會發生在馬
(C) 口蹄疫病毒因血管炎，導致炎症滲出液蓄積於表皮細胞
(D) 豬水泡病（swine vesicular disease）為高傳染性及高致死性傳染病

Seneca Valley Virus (SVV)

Introduction

- An **emerging** swine **vesicular** disease which has been evolved and spread quickly and globally.
- First identified incidentally as a contaminant in cell culture medium during cultivation of PER.C6 cells in 2002.

immortalizing primary human retinal cells

- Sequence analysis revealed that SVV has been circulating in pig herds in the US since 1988 and probably arose in 1985.
- SVV is an **oncolytic** virotherapy candidate for human cancers.

Etiology

Seneca valley virus (senecavirus A)

- Genus *Senecavirus*, family of *Picornaviridae*
- **Non-enveloped** (VP1-4), icosahedral, ~30 nm
- ss, positive-sense RNA genome, ~7.3 kb in length
- **A single serotype, SVV-1**
- Swine is the only natural host.
- Not zoonotic

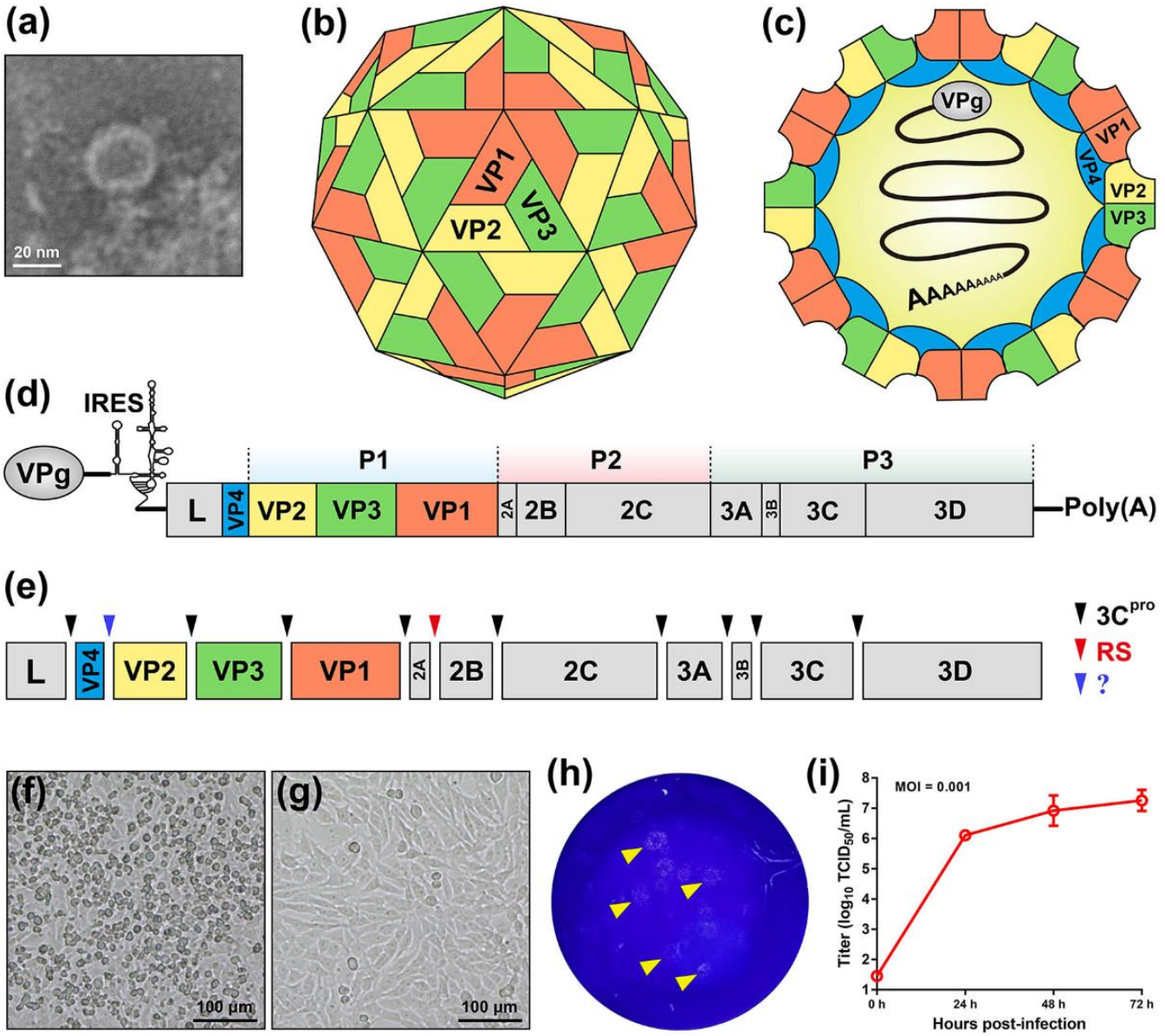
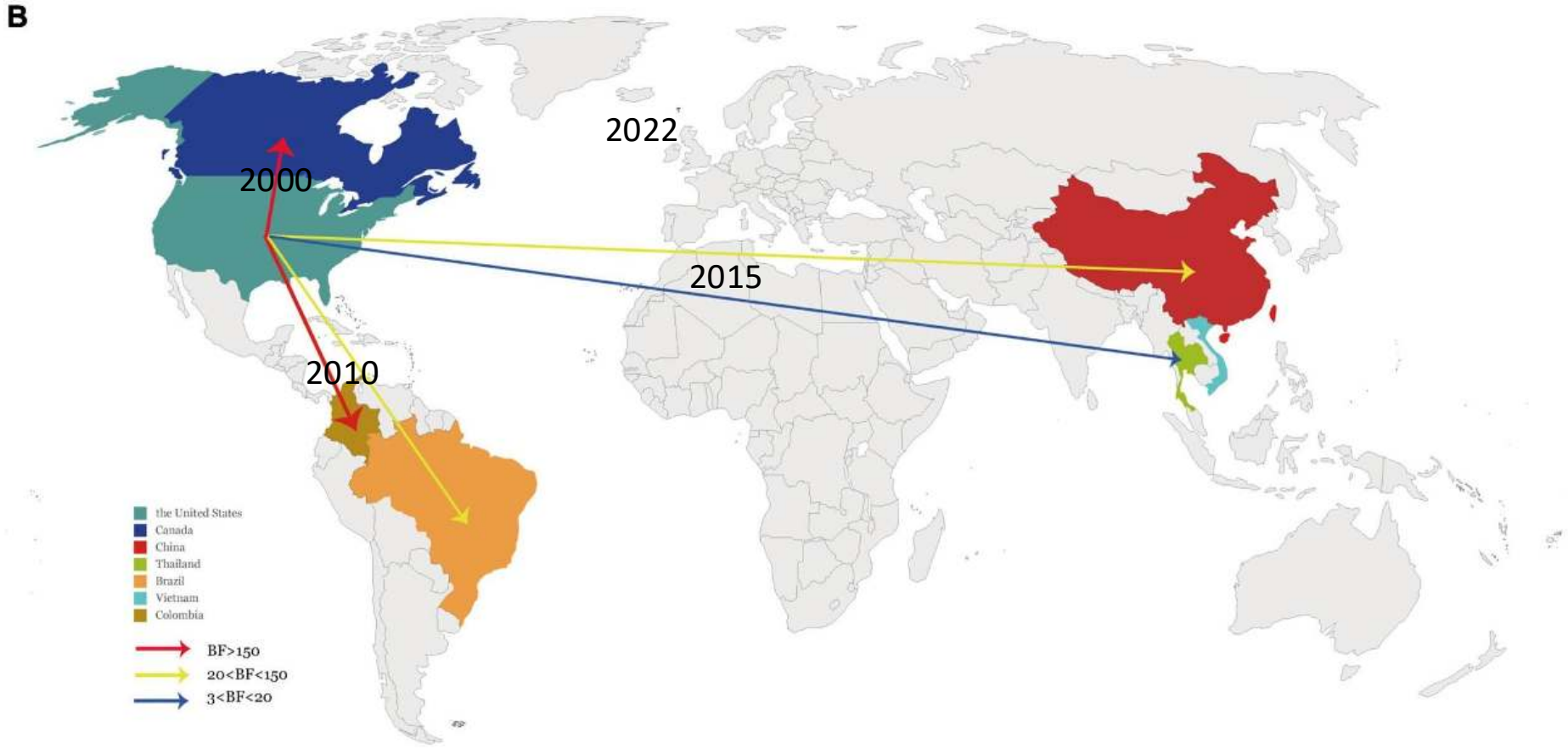


FIGURE 1 | Characteristics of Senecavirus A. Transmission electron microscopy of purified SVA virion **(a)**. Schematic representations of SVA virion: front **(b)** and section **(c)** views. Schematic representation of SVA genome **(d)**. Cleavage sites (arrow-headed) within SVA polyprotein **(e)**. Typical cytopathic effect, cell rounding, at 24 h post infection with SVA **(f)**, and non-infected cell monolayer as mock **(g)**. Plaque formation (arrow-headed) of SVA-infected cell monolayer **(h)**. Multi-step growth curve (MOI = 0.001) of rescued SVA CH-LX-01-2016 cultivated in BSR-T7/5 cells **(i)**. IRES, internal ribosome entry site; $3C^{pro}$, 3C protease; RS, ribosomal skipping; ?, unknown.

Liu et al., 2020

Epidemiology



Modified from Gao et al., 2022

Transmission

- The transmission route has not yet been proven.
- Potential routes include **direct contact** and **fecal-oral transmission**.
- One study showed that **mice** can carry viable SVV (small intestine and feces) and that flies can carry the virus' nucleic acids.

Pathogenesis

- Little is known about SVV entry process.
- **Anthrax toxin receptor 1** (ANTXR1) is the cellular receptor and mediates uncoating in late endosomes.
- No known viral virulence factor has been identified.

Clinical Presentation

- Short incubation time
- **Generally high morbidity** but **low mortality** except for piglets < 1w
- Clinical signs and vesicular lesions cannot be distinguished from those seen in the FMDV infection

  -- **Lameness** and lethargy

  -- Fever

  -- **Vesicles** and erosions on the coronary bands and snouts

  -- Diarrhea, weakness and increased morality in neonatal piglets

Lesions

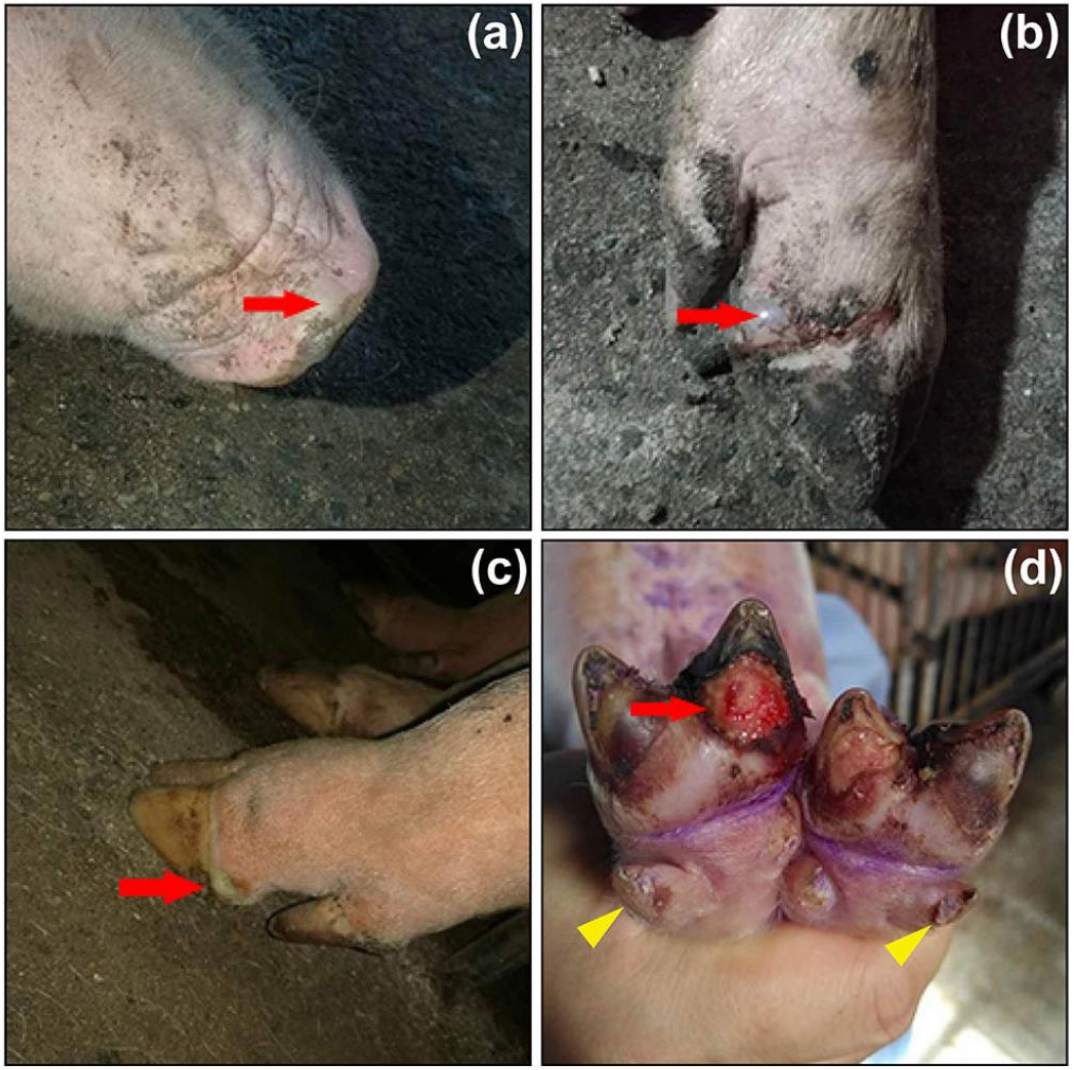


FIGURE 2 | Clinical signs of pigs infected with Senecavirus A in Fujian Province of China in 2017. Fluid-filled vesicles on snout (**a**, red arrow) and coronary bands (**b,c**, red arrows). Ruptured vesicle with ulceration and erosion beneath the toe (**d**, red arrow), and lesions on dewclaws (**d**, yellow arrowheads). Adapted from the reference (23) with permissions of the Elsevier (License No.: 4892830654384).

Lesions

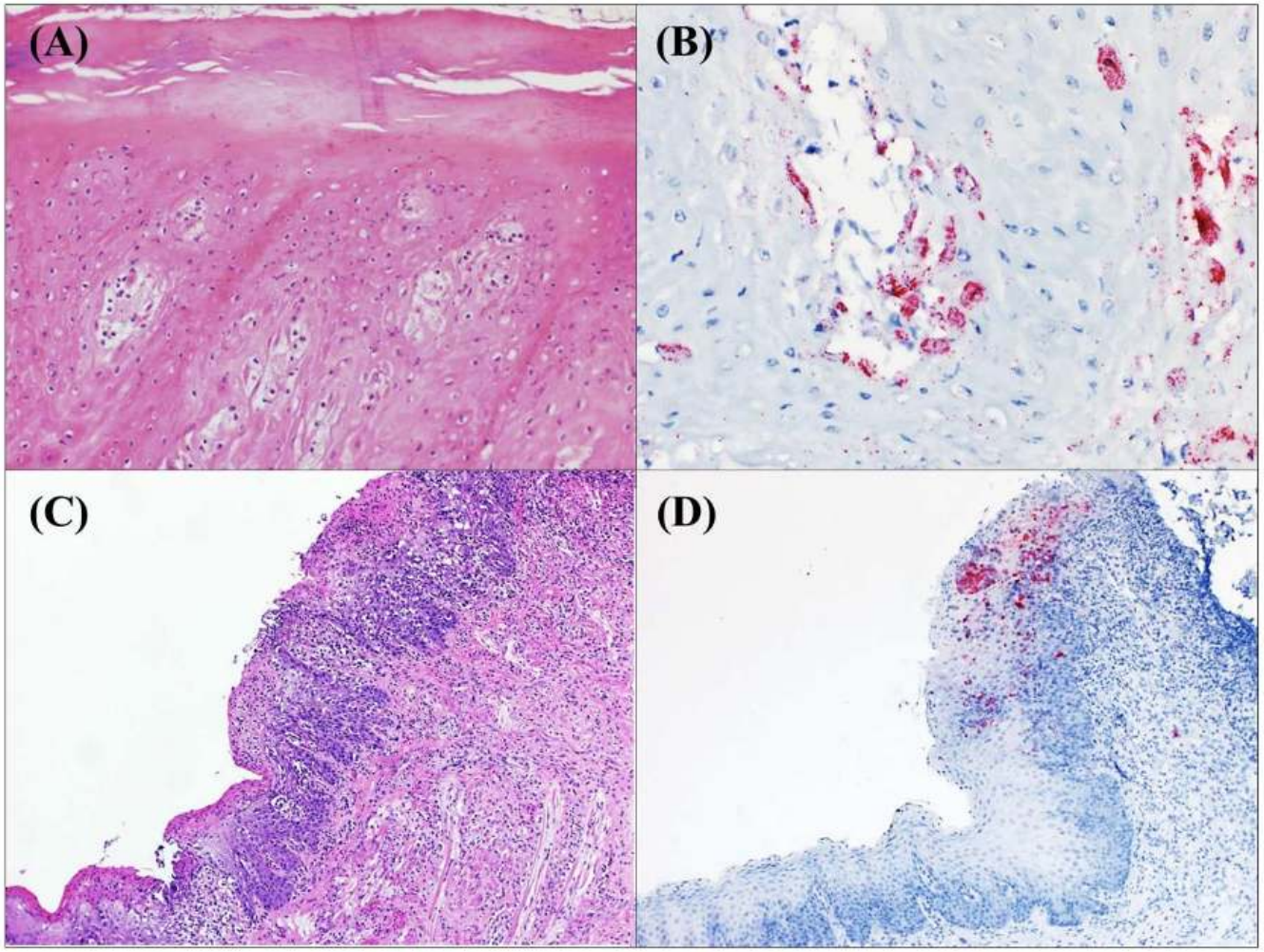


Fig 1. SVV distribution in vesicular lesions in the snout of an affected sow and in necrotizing lesions in tongue of an affected piglet. a) Skin from infected sow. Ballooning degeneration (intracellular edema) of keratinocytes in the *stratum spinosum* with formation of intraepidermal vesicles. H&E, 20x; b) Skin from infected sow. Intraepidermal vesicle showing strong SVV positive staining within the cytoplasm of keratinocytes. ISH-RNA, 40x; c) Tongue from infected piglet. Necrotizing glossitis. H&E, 20x; d) Tongue from the infected piglet shown in (c). Red dots and clusters represent the presence of SVV mRNA within an erosive lesion. ISH-RNA, 40x.

Differential Diagnosis

	Vesicular stomatitis	Vesicular Exanthema of Swine	Foot-and-mouth disease	Swine Vesicular Disease
Virus	Rhabdoviridae	Caliciviridae	Picornaviridae	Picornaviridae
Natural host	Horse, cattle, swine	Swine	cloven-hoofed animals	Swine
Endemic region	America	Eradicated	Africa and Asia	Europe and Asia (including Taiwan)

- Vesicular disease in pigs **must be assumed to be FMD** until proven otherwise.

Laboratory Diagnosis

- Virus detection:
  -- Virion: EM, virus isolation
  -- Antigen: ELISA, IHC in tissue section
  -- Nucleic acids: RT-PCR, RT-qPCR, ISH in tissue section

- Serological test:
  -- Viral neutralization
  -- ELISA for Ab detection

Prevention & Control

No commercial vaccines available against SVV.

- Proper preventative measure, ex strict biosecurity
- Routine surveillance

Case in Taiwan

畜衛所訊 第5期

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電話：02-26212111 傳真：02-26225345
網址：http://www.nvri.gov.tw/index.aspx

花蓮縣豬場出現我國首例矽尼卡谷病毒感染症

花蓮縣動植物防疫所於101年12月22日通報本所，縣轄內有豬場豬隻發現於鼻鏡與蹄冠部出現疑似水泡病徵，花蓮縣動植物防疫所派員將採集之豬隻檢體於12月23日上午送達本所。豬隻檢體（含痂皮及拭子）經RT-PCR檢測結果，口蹄疫病毒(Foot and Mouth Disease Virus；FMDV)及豬水疱病病毒（Swine Vesicular Disease Virus；SVDV）均呈陰性，另血清檢體之口蹄疫非結構蛋白（Non Structural Protein；NSP）抗體檢測結果也呈陰性。病毒分離所得病毒株經Ag-ELISA檢測結果，FMDV及SVDV均呈陰性，另依電子顯微鏡形態學觀察該分離株病毒屬於Picornavirus，其RT-PCR產物經定序分析之比對結果為Seneca Valley Virus(SVV)，有研究報告指出SVV與Swine Idiopathic Vesicular Disease(SIVD)syndrome有關，SVV對豬隻之致病性未來仍需進一步研究探討。



圖、花蓮縣動植物防疫所提供患豬鼻吻部水泡病變之照片。

豬矽尼卡谷病毒(Seneca Valley virus)感染症_2012.12

花蓮縣動植物防疫所
行政院農業委員會家畜衛生試驗所

臺大獸醫專業學院 高啓霏 講

豬矽尼卡谷病毒感染症_2012.12

病歷及臨床症狀:

- 花蓮縣某一貫式豬場, 在養 1,342頭
- 12/22電話通報縣防疫所, **將上市肉豬**經特約獸醫師確認蹄部病變, 疑似水疱
- 肉豬舍 8 欄 (7月齡肉豬 164 頭), 45/164 (27%) 走動困難, 鼻鏡、腳蹄病變



圖 2-1、鼻鏡有水疱病灶及結痂。



圖 2-2、腳蹄冠處有水疱、潰瘍及裂蹄症狀。

組織病變(一)：

■鼻鏡：角質層增厚、角化過度、角化不全、局部靡爛、附著壞死細胞碎屑

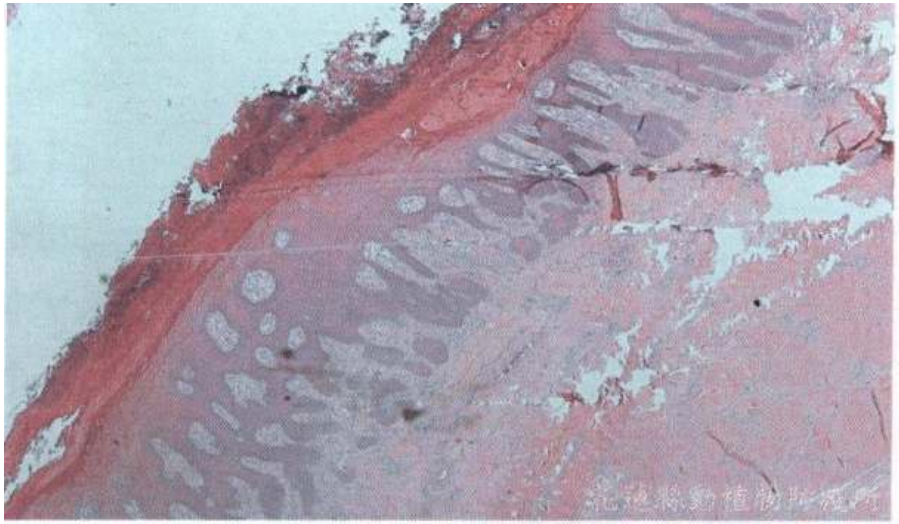


圖 2-3、鼻吻鏡，表皮角質層角化過度、角化不全、局部糜爛(Erosion)並有壞死細胞碎屑附著。表皮層棘細胞增生、皮釘深入真皮層。(H&E stain；40X)

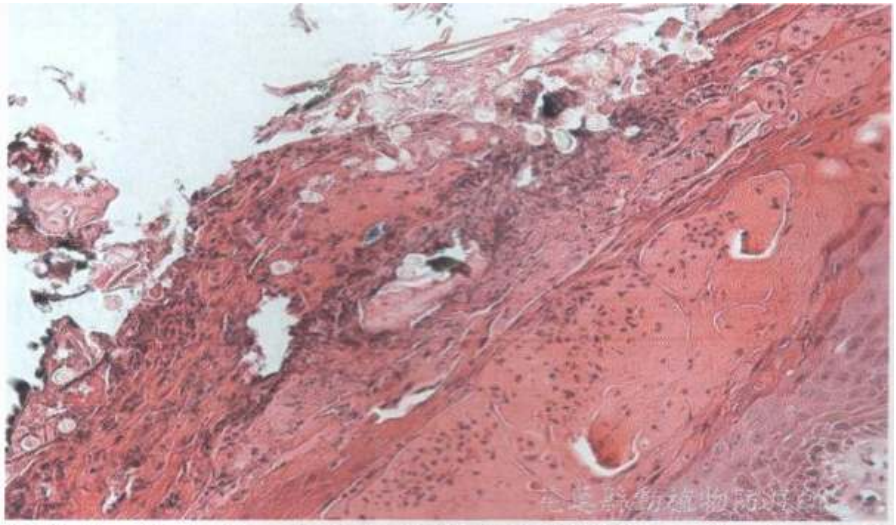


圖 2-4、鼻吻鏡，表皮角質層局部糜爛(Erosion)、角化不全並有壞死細胞碎屑附著。(H&E stain；400X)

組織病變(二)：

■蹄冠：表皮外層組織壞死、棘細胞增生腫脹及空泡化並形成水泡、皮釘深入真皮層、局部糜爛、附著壞死細胞碎屑

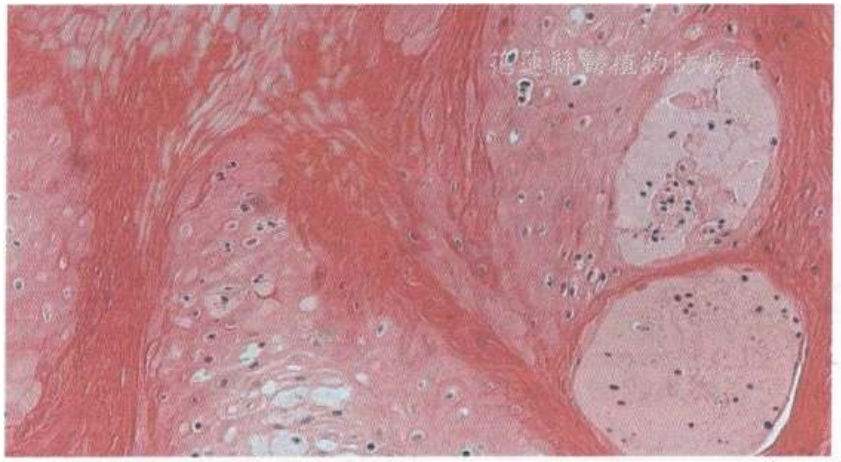


圖 2-7、蹄冠、橫切面，棘細胞腫脹、空泡化、核濃縮與消失，壞死區內死亡之棘細胞融合成一片，形成水泡。（H&E stain；400X）

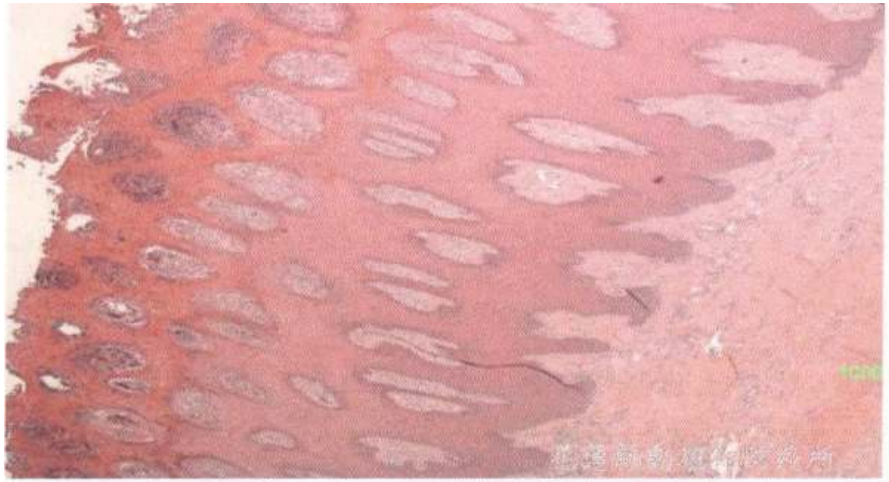


圖 2-5、蹄冠皮膚、橫切面，表皮層外層為壞死組織碎片，棘細胞增生、皮釘深入真皮層。（H&E stain；40X）

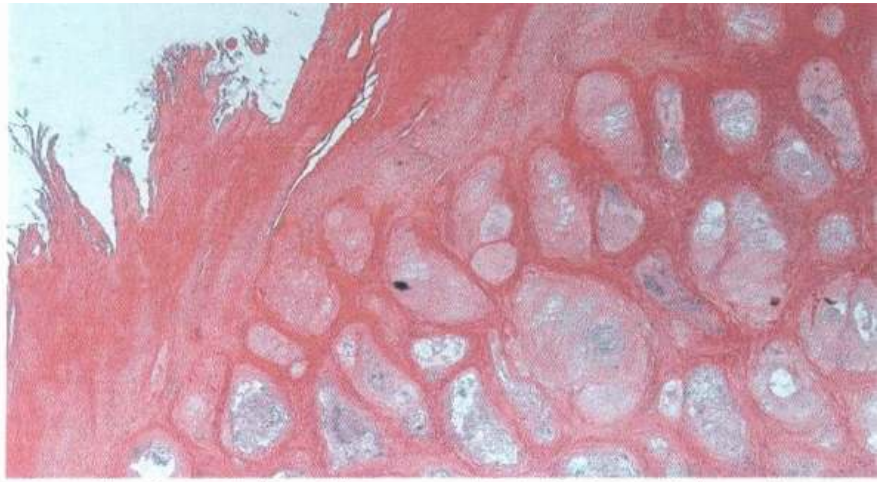


圖 2-6、蹄冠皮膚、橫切面，表皮層角化過度、棘細胞腫脹、空泡化、核濃縮與消失。（H&E stain；40X）

實驗室診斷(一)：

- 血清學診斷：ELISA 抗體 - FMD與SVD 陰性
- 分子生物學檢測： RT-PCR - FMD與SVD 陰性；水疱性口炎(VS)與豬水疱疹(VES) 陰性
- 病毒增殖與分離 - EBK增殖之分離病毒
  - Ag-ELISA：FMD與SVD陰性
  - 電子顯微鏡檢查：無封套正二十面體顆粒，大小為25 nm ***(Picornavirus)***。

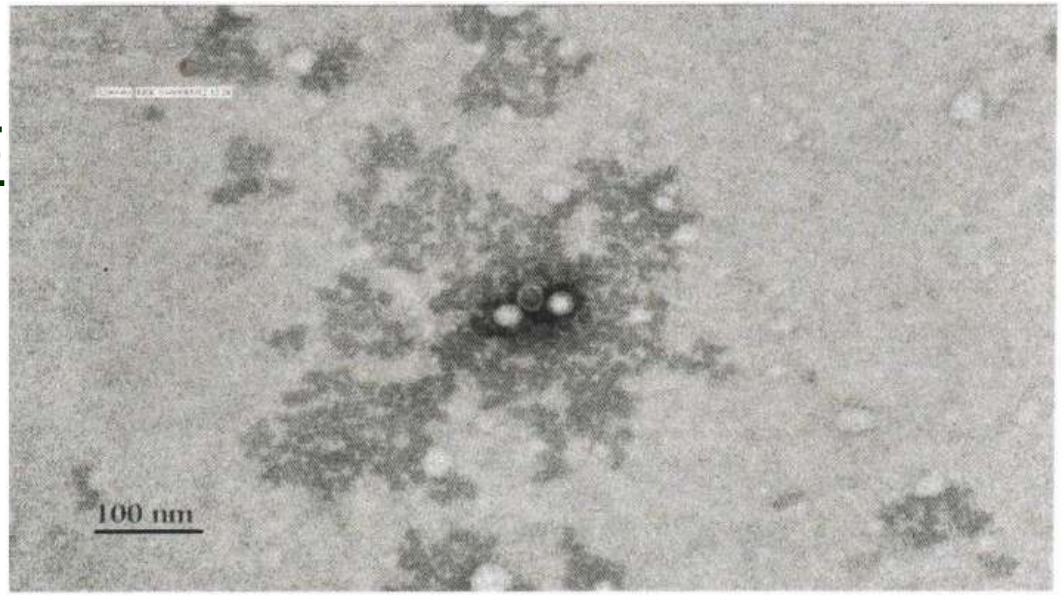


圖 2-9、自 EBK 細胞所分離之病毒，以電子顯微鏡觀察病毒形態，為無封套之正二十面體病毒顆粒。(家衛所提供)

實驗室診斷(二)：

Embryonal bovine kidney cells

- 病毒增殖分離 - EBK 增殖之分離病毒 (續)
  - 病毒核酸 (RT-)PCR：FMD, SVD 陰性；VSV 陰性；VESV 所得非預期核酸片段定序，再經序列比對，發現**與 Seneca valley virus (SVV)序列相似度 98%**。

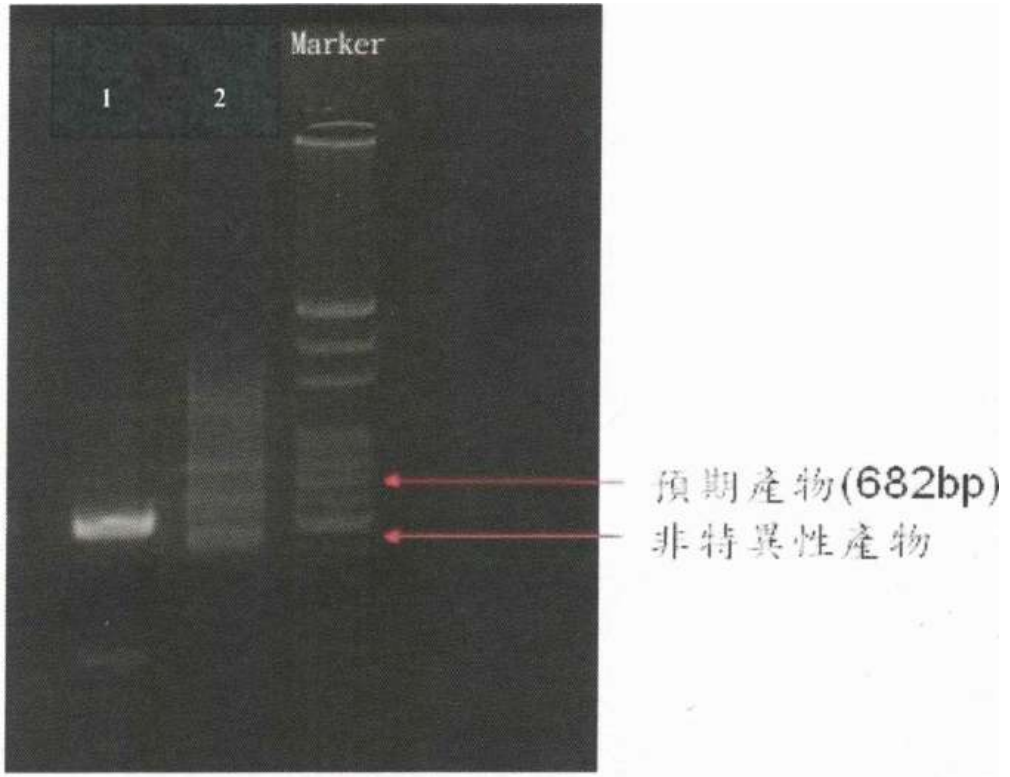


圖 2-8、編號 1 自 EBK 細胞所分離之病毒，以 VESV 特異性引子增幅出非預期核酸片段的非特異性產物，與原 VESV 引子設計增幅之 682bp 片段大小不同。(家衛所提供)

疫情控制：

- **12/22**採樣送驗，移動管制，依「口蹄疫防疫作業標準」對該場疑患及同欄所有豬隻**當日**採**預防性撲殺** - 4 欄 107 頭肉豬 (45頭有水疱性病灶)
- **12/23** 凌晨完成撲殺作業，繼完成全場補強一劑口蹄疫疫苗，抽測場內免疫口蹄疫疫苗後 3-5 週齡豬隻血清 (15頭) 送檢口蹄疫抗體
- 環境消毒，半徑 3 公里內飼養偶蹄類動物場訪視調查 (結果皆無異狀)
- **12/24-28** 縣防疫所人員每日訪視、消毒車輛進行環境消毒，該場其它豬隻皆無異狀

後續:

- 為台灣首例矽尼卡谷病毒 (SVV) 感染症；為新興的豬自發性水疱性疾病，臨床症狀與所知四種豬隻水疱性疾病無法區分，需仰賴實驗室診斷
- **SVV僅發現於成豬**，並無出生哺乳豬或仔豬感染病例，若為單純 SVV 感染，**流行病學調查之致死率為 0%**
- SVV 自然宿主及致病機轉尚不明；有報告指出可**藉由餵飼被鼠科動物污染飼料**而感染，並可經由運輸車輛而散播 → 推測本案例之傳播源為老鼠，針對此防治之
- Non-OIE Listing Disease

影響：

- 將 SVV 疫病資訊納入水疱性疾病之基層防疫宣導
- 行政院農業委員會動植物防病檢疫局 (防檢局) 著手修訂「口蹄疫防疫作業標準」，關於水疱性病原之實驗室診斷鑑定程序

鞏固國內水疱性疾病防疫網，期能早日成為非疫區

- 2018年4月上旬(清明節)台南肉品市場繫留欄，兩隻肉豬出現水疱臨床症狀，採樣送畜衛所，VI & 分生診斷為台灣第二例 SVV

家畜衛試所研報No.42：61~70(2007)

2006 年送檢豬隻病材病毒性疾病檢驗結果

黃天祥*、鄧明中、王 羣、張家宜、李淑慧、潘居祥、黃有良、李敏旭、鍾明華

行政院農業委員會家畜衛生試驗所豬瘟研究組

臺北縣淡水鎮中正路 376 號

摘要 2006 年共接獲縣市家畜疾病防治所或動物防疫所送檢 76 場次 211 頭豬隻臟器檢體。經接種於 PK-15 等細胞進行病毒分離檢測結果，以第二型豬環狀病毒（porcine circovirus 2; PCV2） 有 20 場病毒分離呈陽性為最多；其次依序為豬鐵士古病毒（porcine teschovirus ; PTV）有 19 場，豬腸病毒有 12 場（其中 porcine enterovirus 8; PEV8 有 9 場，porcine enterovirus 9; PEV9 有 3 場），豬生殖與呼吸綜合症病毒（porcine reproductive and respiratory syndrome virus; PRRSV）和里奧病毒（reovirus；Reo V）各有 5 場，假性狂犬病病毒（pseudorabies virus；PrV）有 3 場，豬流感病毒 （swine influenza virus；SIV；H1N2 ）、豬星狀病毒病毒(porcine astrovirus；PAstV) 和兔化豬瘟疫苗毒 (lapinized hog cholera virus；LPCV) 各有 1 場病毒分離呈現陽性。此外，由屏東送檢之運豬車血水首次分離出矽尼卡谷病毒 (seneca valley virus；SVV)。